A Probabilistic Analysis of the Conditioning of Algebraic Rootfinding Methods

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Abstract

To be written.

1 Introduction

1.1 Overview of the main contributions

The results of [18] and [16] say that every known algebraic rootfinder is exponentially unstable in the worst case: there exist systems for which the intermediate subproblem (an eigenvalue problem or a univariate polynomial) is worse conditioned than the original root by a factor $\sigma^{-(d-1)}$, or even σ^{-2^d+2} for elimination. These statements rest on carefully designed examples. This paper asks what happens for typical systems. We model "typical" by i.i.d. complex Gaussian coefficients with fixed degree n in each of the d variables, the same degree model chosen in [18], and we study the expected value of the logarithm of the conditioning ratio, averaged over all roots of the system. Probabilistic analysis of this kind has long been central to the other main family of polynomial system solvers, homotopy continuation methods: the average-case programme initiated by Shub and Smale [19], and systematized in the monographs of Blum, Cucker, Shub, and Smale [6] and of Bürgisser and Cucker [10], led, through the probabilistic solution of Smale's seventeenth problem by Beltrán and Pardo [2] and the deterministic near-solution of Bürgisser and Cucker [9], to Lairez's deterministic average polynomial-time algorithm [17]. The present paper brings the same average-case viewpoint to the algebraic rootfinders.

The first outcome is structural. For every method in the family, the ratio at a root factorizes into a *local* part, depending only on the root and on the system near it, and a *structural* part, encoding how the method couples distinct roots. The local part is the same for all methods, and we compute it exactly: it equals $(d+1)\log n$ up to an additive $O(d)$. It should be read as a change of units between "conditioning of the root" and "conditioning of the subproblem", not as an instability; a scale-free version of the comparison removes it entirely (Remark 11.1).

The second outcome is a hierarchy. The structural parts fall into three classes. For the hidden-variable resultant methods in their canonical Cayley and Sylvester forms there is none: an exact identity of [18] makes the ratio a purely local quantity, and for Cayley we obtain a complete and unconditional average-case theorem, verified numerically, with the conclusion that the exponential worst-case instability of [18] is a tail event, not the typical behaviour; in the relative metric the method is, on average, neutral with respect to the root conditioning. Multiparameter eigenvalue problems are also purely local. For Gröbner elimination and the rational univariate representation, the structural part is a logarithmic energy of pairs of roots, which requires only the two-point statistics of the root set. For normal form and Macaulay methods it is the eigenvector conditioning of the multiplication matrix, a quantity involving all roots at once. We prove that this hardest term is finite for every d, and the numerical protocol, executed at $d = 1, 2, 3$, shows that it does not follow the random-matrix calibration at $d = 1$ and 2, where it appears bounded and the multiplication matrix is far more rigid than a Ginibre matrix; at $d = 3$ the measured overlaps run parallel to the Ginibre control, a constant below it, up to $N = 1296$: the third dimension appears to sit in the growth regime at accessible sizes. If so, the average loss of the normal-form method is fully described by the local terms; the pairwise-class terms are computed at leading order: one is an outer-root effect, the other a closed-form logarithmic energy. Section 12 converts everything to the relative metric, the one in which conclusions should be drawn, and the family separates cleanly there: only the rational univariate representation pays an exponential average penalty, with a closed-form constant. The message is that the exponential instability of the worst case is produced by atypical configurations, chiefly nearly colliding roots; for random systems the expected loss is modest, in agreement with our earlier numerical evidence.

A byproduct of independent interest is what we call the outer-root phenomenon. For Kac-type random systems, logarithmic averages over the roots are not dominated by the roots near the unit torus. About half of the roots lie outside, at rescaled distances with a heavy $1/u$ tail, and the local scales of the system at such roots are exponentially large. Their total contribution to every "average of a logarithm" is of the same order as that of the bulk. Several natural heuristics fail at leading order for exactly this reason. The Kostlan ensemble exhibits the same effect in extreme form: its root measure does not concentrate at all, the affine average ratio grows linearly in n, and the projectively invariant normalization is identified as the only sensible yardstick there. The total-degree model, by contrast, reproduces the separate-degree coefficients exactly, through a different mechanism: its heavy tail is carried by roots receding radially in all coordinates at once.

All local components are established rigorously, as is the finiteness of every structural term, and the overlap term is proved bounded at $d = 1$ and $d = 2$; what remains open is the sharp growth of the full-configuration terms, together with the rigor items collected in Section 14.

1.2 The quantity of interest

Let $P = (p_1, \dots, p_d) : \mathbb{C}^d \to \mathbb{C}^d$ be a square polynomial system with N simple roots $z_1, \dots, z_N$. An algebraic rootfinder computes the roots through an intermediate problem: normal-form methods diagonalize a multiplication matrix on the quotient ring; hidden-variable resultant methods solve $\det R(x_1) = 0$ for a resultant matrix polynomial; Gröbner elimination and the rational univariate representation produce a univariate polynomial; multiparameter eigenvalue methods and the Macaulay resultant produce generalized eigenvalue problems. Each method M has a condition number $\kappa_{\mathrm{M}}(x^*)$ for the subproblem it constructs at a root, defined in [16, 18]. The intrinsic condition number of the root is $\kappa_{\mathrm{root}}(x^*) := \|J(x^*)^{-1}\|_2 = 1/\sigma_{\min}(J(x^*))$, with J the Jacobian. The object of study is the average logarithmic conditioning ratio

$$\mathcal{R}_n^{\mathrm{M}} := \frac{1}{N}\,\mathbb{E}\Big[\sum_{j=1}^{N} \log \frac{\kappa_{\mathrm{M}}(z_j)}{\kappa_{\mathrm{root}}(z_j)}\Big], \tag{1}$$

where the expectation is over the random coefficients (Section 2) and the sum runs over all roots. We use $\mathbb{E}[\log]$ rather than $\log \mathbb{E}$ because it measures the typical number of digits lost, it is additive under the factorizations below, and it is robust to the heavy tails that condition numbers have.

All the methods above are treated, and the plan rests on one structural fact, proved case by case below: for every method, the ratio in (1) factors at each root into a *local* part, a function of the root and of the system near it, and a *structural* part, encoding how the method couples distinct roots. The local part turns out to be the same for all methods, so it is computed once, in Sections 4–8; the methods differ only in their structural parts, which are classified and analysed in Sections 9 and 10. The common machinery is developed first, and among the structural parts the normal form's is treated before the others, for two reasons. First, its condition number, by [16, Thm. 6.2],

$$\kappa_{\mathrm{eig}}(x^*) = \frac{\|[\det Q]_{\mathcal{B}}\|_2\, \|\mathcal{B}(x^*)\|_2}{|\det J(x^*)|}\,(1 + |\lambda|), \tag{2}$$

where $\mathcal{B}$ is a monomial basis of the quotient ring R/I, Q is any matrix polynomial with $P(x) = Q(x)(x - x^*)$, and $\lambda = g(x^*)$ is the eigenvalue of the multiplication matrix M_g associated with the root (g a fixed generic polynomial such as $g = x_1$), leads to the first of the two exact reductions on which everything rests. Second, its structural part is the only genuinely global one, the eigenvector conditioning of M_g, so it is the case that needs the full machinery; the other methods are then treated on the same footing, at little extra cost, and some, like the Cayley resultant, end up completely solved. We write $\mathcal{R}_n := \mathcal{R}_n^{\mathrm{eig}}$ for the normal-form instance.

To avoid any ambiguity about which condition numbers enter (1): for the normal form we take (2), the exact formula of [16, Thm. 6.2], as the definition; for the hidden-variable resultants we use the eigenvalue condition number of the resultant matrix polynomial as defined in [18, Def. 2.10], for which [18, Lem. 2.11] is an equality; for Gröbner elimination

and the rational univariate representation we use the conditioning $1/|g'|$ of the univariate subproblem, as in [16]. All of these are actual condition numbers, not bounds. The single exception is the Macaulay method, for which [16, Thm. 7.1] provides only a lower bound; the corresponding statements in Section 10.8 are lower bounds and are marked as such.

1.3 Root averages carry a Jacobian weight

One identity organizes the whole computation. For any measurable quantity $\varphi(X; P)$ depending on a point and on the system,

$$\mathbb{E}\Big[\sum_{z:\,P(z)=0}\varphi(z;P)\Big] = \int_{\mathbb{C}^d}\mathbb{E}\big[\varphi(X;P)\,|\det J(X)|^2 \;\big|\; P(X)=0\big]\,p_{P(X)}(0)\,dX, \tag{3}$$

where $p_{P(X)}(0)$ is the value at the origin of the probability density of the Gaussian vector $P(X) \in \mathbb{C}^d$. This is a change of variables (the coarea formula), not a probabilistic subtlety; the factor $|\det J|^2$ is the real Jacobian determinant of the map $\mathbb{C}^d \to \mathbb{C}^d$, and the square comes from holomorphy. Identity (3) is, in other words, the Kac–Rice formula for the zero set of the Gaussian field P, in the weighted (or marked) form in which the summand φ may depend on the whole field, specialized to the holomorphic case: the Kac–Rice weight $|\det DF|$ of a real field F becomes $|\det J|^2$, which is sign-definite and polynomial in the field and its derivatives, so none of the absolute-value smoothness issues of the real theory arise. We refer to Azaïs–Wschebor [1, Ch. 6] for the general theory, and to Berzin–Latour–León [4] for a contemporary overview of the formula's variants and applications. The practical content is that "conditioning on a root at X" is not the naive conditioning on $\{P(X) = 0\}$: the correct conditional law – in point-process language, the Palm distribution of the root set – carries the weight $|\det J(X)|^2$, which down-weights badly conditioned roots. Forgetting the weight gives wrong answers; keeping it turns out to simplify the computations, because the weighted Gaussian ensembles that appear are again classical (Section 7). The same mechanism underlies the moment computations of Shub and Smale [19] and, at $d = 1$, the root-density computations of Edelman and Kostlan [14]. Hypotheses and an extension to unbounded φ are checked in Section 6, where the identity is deduced from the Rice formulas of [1, Thms. 6.2 and 6.4].

1.4 Results

Items (a)–(c) below are common to all methods; (d) and (e) concern the structural part of the normal form; (f) the other methods.

(a) *Reduction I* (Section 3, exact). In R/I one has $[\det Q] = \det J(z_j)\,[\ell_j]$, with ℓ_j the Lagrange idempotent at z_j. Hence $\kappa_{\mathrm{eig}}(z_j) = \Lambda_j\,(1 + |\lambda_j|)$, where $\Lambda_j \geq 1$ is exactly the Wilkinson condition number of the eigenvalue λ_j of M_g, and

$$\frac{\kappa_{\mathrm{eig}}(z_j)}{\kappa_{\mathrm{root}}(z_j)} = \sigma_{\min}(J(z_j)) \cdot \Lambda_j \cdot (1 + |\lambda_j|).$$

The reduction of $\det Q$ modulo I, which had appeared to be the main obstacle, disappears from the problem.

(b) *Reduction II* (Sections 4–7, exact). Conditionally on $P(X) = 0$, the Jacobian is $G\,\Sigma(X)^{1/2}$ with G a $d \times d$ matrix of i.i.d. standard complex Gaussians and $\Sigma(X)$ explicit and diagonal in our model. The weight $|\det G|^2$ maps the singular values of G to those of a $(d+1) \times d$ Gaussian matrix. Every local root average is thereby an explicit integral.

(c) *Local terms* (Section 8). With $A_n := \frac{1}{N}\mathbb{E}\sum_j \log \sigma_{\min}(J(z_j))$ and $C_n := \frac{1}{N}\mathbb{E}\sum_j \log(1 + |\lambda_j|)$,

$$A_n = (d+1)\log n + O(d), \qquad C_n = \log 2 + o(1).$$

The computation requires the exact law of the root radii at scale $1/n$ (Proposition 8.1) and reveals the outer-root phenomenon (Section 8.2).

(d) *The structural term* (Section 9). Everything nonlocal sits in $\Theta_n := \frac{1}{N}\mathbb{E}\sum_j \log \Lambda_j \geq 0$. We give exact identities, prove $\Theta_n < \infty$ for every d (Proposition 9.4), identify the mechanism of its tails, prove $\Theta_n = O(1)$ at $d = 1$ and $d = 2$ (Theorem 9.1), and execute the numerical test at $d = 1, 2, 3$: at $d = 3$ the overlaps run parallel to the Ginibre control, leaving the regime open there.

(e) *Main theorem* (Section 11, conditional):

$$\mathcal{R}_n = (d+1)\log n + \Theta_n + O(d), \qquad 0 \leq \Theta_n < \infty,$$

and, by Theorem 9.1, $\mathcal{R}_n = (d+1)\log n + O(1)$ unconditionally at $d = 1, 2$; the regime at $d \geq 3$ is open.

(f) *The full family* (Section 10). The other methods of [16] obey the same skeleton with structural terms classified as: none (hidden-variable resultants in their canonical Cayley and Sylvester forms, by an exact identity from [18]; multiparameter eigenproblems), pair energies (Gröbner elimination, rational univariate representation), full-configuration overlaps (Macaulay). For the Cayley method this yields an unconditional theorem: the average ratio equals $-\frac{d(d-1)}{2}\log n + O_d(1)$ (Theorem 10.1), confirmed numerically at $d = 2$ down to the finite-n constants (Section 10.3); in the absolute metric of [18] the eigenproblem is typically *better* conditioned than the root, a units effect that disappears in the relative metric, where the average ratio is neutral (Section 10.4).

1.5 Notation

$\|\cdot\|_2$ denotes the Euclidean norm of a vector and the spectral norm of a matrix, $\|\cdot\|_F$ the Frobenius norm, and $\sigma_1 \geq \cdots \geq \sigma_{\min}$ the singular values. G always denotes a $d \times d$ matrix with i.i.d. $\mathbb{C}\mathcal{N}(0,1)$ entries, and $\mathbb{E}_W[\cdot] := \mathbb{E}[\cdot\,|\det G|^2]/d!$ the associated weighted expectation (Section 7). The roots of the system are $z_1, \ldots, z_N$ with $N = d!\,n^d$; $\mathcal{S}$ is a

monomial staircase, $\mathcal{B}$ the corresponding basis of R/I, $V = (z_j^\beta)_{\beta\in\mathcal{S},\,j}$ the evaluation matrix with columns v_j, ℓ_j the Lagrange idempotents with coordinate vectors $V^{-T}e_j$, and $\Lambda_j := \|V^{-T}e_j\|_2\,\|Ve_j\|_2$. $\psi = \Gamma'/\Gamma$ is the digamma function, $H_d = \sum_{k=1}^{d} 1/k$, γ Euler's constant, and $k_m(r) := \sum_{j=0}^{m} r^j$. Following [18], Ω denotes the closed unit polydisk $\bar{\mathbb{D}}^d$, the reference domain of the rootfinding problem, so that $\sup_\Omega |p_i|$ is the sup-norm of p_i there (equal to the sup on the torus $\mathbb{T}^d$ by the maximum principle). Asymptotic notation $O_d(\cdot)$ indicates constants depending on d only; all logarithms are natural.

2 Model and probabilistic background

Definition 2.1 (separate-degree i.i.d. model) *Fix $d \geq 1$ and $n \geq 1$. For $i = 1, \ldots, d$,*

$$p_i(x) = \sum_{\alpha\in\{0,\ldots,n\}^d} c_{i,\alpha}\, x^\alpha, \qquad c_{i,\alpha} \sim \mathbb{CN}(0,1) \text{ i.i.d.},$$

where $x^\alpha := \prod_l x_l^{\alpha_l}$ and $\mathbb{CN}(0,1)$ has independent real and imaginary parts of variance $\frac{1}{2}$.

This is the maximal-degree model of [18], where it is motivated by its link to tensor-product constructions [18, Sec. 2.1]. A further reason to prefer it here: fixing the maximal degree in each variable, rather than the total degree, keeps the Newton polytopes equal to boxes, for which the resultant constructions of Section 10 are generically nonsingular and there are generically no roots at infinity. Each p_i is a Gaussian random function with covariance kernel

$$K(x,y) := \mathbb{E}\big[p_i(x)\overline{p_i(y)}\big] = \prod_{l=1}^{d} k_n(x_l\bar{y}_l), \qquad k_n(t) := \sum_{j=0}^{n} t^j, \tag{4}$$

and the d functions are independent. Write $S(X) := K(X,X) \geq 1$ and $r_l = |X_l|^2$.

By the Bernstein–Kushnirenko–Khovanskii (BKK) theorem [3], with the normalization $\mathrm{MV}(Q,\ldots,Q) = d!\,\mathrm{Vol}(Q)$, generic systems with box supports have

$$N = d!\ \mathrm{Vol}([0,n]^d) = d!\, n^d \tag{5}$$

simple roots in $(\mathbb{C}^*)^d$; since all d supports coincide, this is in fact the unmixed case due to Kushnirenko. For $d = 2$ this is the bidegree count $2n^2$. Bernstein's equality criterion [3] makes the genericity explicit: the count (5) is attained unless some facial subsystem has a root in the torus, and each such failure is a proper algebraic condition on the coefficients. Since the coefficient law has a density and the degenerate systems thus form a proper algebraic subset, almost surely all N roots are simple, the ideal $I = \langle p_1, \ldots, p_d\rangle$ is radical and zero-dimensional, and evaluation at the roots identifies R/I with $\mathbb{C}^N$. We fix a monomial basis $\mathcal{B} = \{x^\beta : \beta \in \mathcal{S}\}$ of R/I (from a Gröbner or border basis for a fixed order), and set

$$V \in \mathbb{C}^{N\times N}, \quad V_{\beta j} = z_j^\beta, \qquad v_j := Ve_j = \mathcal{B}(z_j).$$

The Lagrange idempotents $\ell_j \in R/I$ satisfy $\ell_j(z_k) = \delta_{jk}$ and have coordinate vectors $[\ell_j]_\mathcal{B} = V^{-T}e_j$.

2.1 Probabilistic background in brief

Four tools are used, none deeper than linear algebra plus calculus.

(1) *Counting formula.* Identity (3) converts a sum over roots into an integral, exactly as one converts a sum over eigenvalues into a contour integral. The only probabilistic inputs are the density of $P(X)$ at 0 and a conditional expectation.

(2) *Gaussian conditioning is least squares.* If (u, v) is jointly Gaussian, the law of u given $v = 0$ is again Gaussian, with covariance the Schur complement of the joint covariance. Applied to $(J(X), P(X))$ this gives Lemma 4.1. No other property of Gaussians is used.

(3) *One random matrix.* G denotes a $d \times d$ matrix with i.i.d. $\mathbb{C}\mathcal{N}(0,1)$ entries. We use two classical facts: $\mathbb{E}|\det G|^2 = d!$ (a two-line computation), and the smallest eigenvalue of G^*G is exponentially distributed with rate d [13], so $\sigma_{\min}(G) \approx d^{-1/2}$ and all moments of $\log \sigma_{\min}(G)$ are finite.

(4) *A change of measure.* The weight $|\det G|^2$ from (3) multiplies the eigenvalue density of G^*G by $\prod_i \lambda_i$. Lemma 7.1 identifies the result with the eigenvalue law of $\widetilde{G}^*\widetilde{G}$ for a $(d+1) \times d$ Gaussian matrix $\widetilde{G}$. So all weighted expectations reduce to unweighted ones for a slightly rectangular matrix.

We also use the digamma function $\psi = \Gamma'/\Gamma$, for which $\psi(k) = -\gamma + \sum_{i=1}^{k-1} \frac{1}{i}$ at integer arguments, and the anti-concentration inequality of Carbery and Wright [11]: if q is a polynomial of degree k in standard Gaussian variables, then $\mathbb{P}\big(|q| \le \varepsilon \, \|q\|_{L^2}\big) \le C\, k\, \varepsilon^{1/k}$, whence $\mathbb{E}\big[-\log(|q|/\|q\|_{L^2})\big] = O(k \log k)$. This replaces, for our purposes, any finer information about the law of q.

3 Reduction I: what the normal-form condition number is

Theorem 3.1 *Almost surely, for every root z_j and every matrix polynomial Q with $P(x) = Q(x)(x - z_j)$,*

$$[\det Q] = \det J(z_j)\,[\ell_j] \qquad \text{in } R/I.$$

In particular $\|[\det Q]_{\mathcal{B}}\|_2 = |\det J(z_j)|\, \|V^{-T} e_j\|_2$, for every choice of Q.

Proof. At a root $z_k \neq z_j$: $Q(z_k)(z_k - z_j) = P(z_k) = 0$ and $z_k - z_j \neq 0$, so $\det Q(z_k) = 0$. At z_j: differentiating $P(x) = Q(x)(x - z_j)$ and evaluating at z_j kills the term containing $(x - z_j)$ and gives $J(z_j) = Q(z_j)$, so $\det Q(z_j) = \det J(z_j)$. The polynomial $\det Q - \det J(z_j)\,\ell_j$ vanishes at all N roots of the radical zero-dimensional ideal I, hence lies in I. □

Corollary 3.1 *With $\Lambda_j := \|V^{-T} e_j\|_2\, \|V e_j\|_2$,*

$$\kappa_{\text{eig}}(z_j) = \Lambda_j\,(1 + |\lambda_j|), \qquad \Lambda_j \ge 1, \qquad \frac{\kappa_{\text{eig}}(z_j)}{\kappa_{\text{root}}(z_j)} = \sigma_{\min}(J(z_j))\,\Lambda_j\,(1 + |\lambda_j|),$$

and Λ_j is the Wilkinson condition number of the eigenvalue λ_j of M_g in the basis $\mathcal{B}$.

Proof. Substituting Theorem 3.1 into (2) cancels $|\det J|$, and $\|\mathcal{B}(z_j)\| = \|Ve_j\|$. The bound $\Lambda_j \geq 1$ is Cauchy–Schwarz on $1 = \ell_j(z_j) = [\ell_j]_{\mathcal{B}}^T \mathcal{B}(z_j)$. For the eigenvector statement: if c is the coordinate vector of $f \in R/I$ then $f(z_j) = v_j^T c$ and $(gf)(z_j) = \lambda_j f(z_j)$, so $M_g^T v_j = \lambda_j v_j$, and from $M_g^T V = V \operatorname{diag}(\lambda)$ one gets $M_g(V^{-T}e_j) = \lambda_j(V^{-T}e_j)$. In the standard convention (right eigenvector v, left eigenvector u with $u^*A = \lambda u^*$, condition number $\|u\|\|v\|/|u^*v|$), for $A = M_g^T$ the left eigenvector is $u_j = \overline{V^{-T}e_j}$; then $u_j^* v_j = e_j^T V^{-1} V e_j = 1$ exactly, $\|u_j\| = \|V^{-T}e_j\|$, and the Wilkinson number equals Λ_j. □

REMARK **3.1** *Three consequences. First, the reduction of* $\det Q$ *modulo* I *is not a separate difficulty: it* is *the eigenvector conditioning, so the two lines of attack pursued in our earlier notes (bounding the reduction; studying* M_g*) coincide. Second, the method can never beat the root conditioning by more than* $\sigma_{\min}(J)(1+|\lambda|)$*, deterministically. Third,* Λ_j *depends on the basis* $\mathcal{B}$*, as it must:* κ_{eig} *describes the algorithm, and the algorithm chooses a basis. The convention* $q = \det Q$ *in* (2) *is confirmed by [16, Thm. 6.2], which also proves the evaluation identity* $[\det Q]_{\mathcal{B}}^T \mathcal{B}(x^*) = \det J(x^*)$*, consistent with Theorem 3.1.*

4 Reduction II, part 1: the Jacobian at a root

Lemma 4.1 *Fix* $X \in \mathbb{C}^d$*. Conditionally on* $P(X) = 0$*, the rows of* $J(X)$ *are i.i.d. centered complex Gaussian vectors with covariance*

$$\Sigma(X) = S(X)\, H(X), \qquad H_{jk}(X) := \frac{\partial^2}{\partial x_j\, \partial \bar{y}_k} \log K(x,y)\Big|_{y=x=X};$$

equivalently $J \overset{d}{=} G\, \Sigma(X)^{1/2}$ *with* G *as in Section 2.1.*

Proof. Rows are independent because the p_i are. For one row, apply Gaussian conditioning (Schur complement) to the jointly Gaussian vector $(p_i(X), \partial_1 p_i(X), \dots, \partial_d p_i(X))$: with $K_j := \partial_{x_j} K|_X$ and $K_{j\bar{k}} := \partial_{x_j}\partial_{\bar{y}_k} K|_X$, the conditional covariance is $K_{j\bar{k}} - K_j \overline{K_k}/S$, which equals $S\, \partial_{x_j}\partial_{\bar{y}_k} \log K$ by expanding the logarithmic derivative. □

Lemma 4.2 *The expected number of roots per unit volume at* X *is*

$$\rho(X) = \frac{d!}{\pi^d}\, \det H(X).$$

Proof. Apply (3) with $\varphi = 1$: $\rho(X) = \mathbb{E}[\,|\det J|^2 \mid P(X) = 0\,]\, p_{P(X)}(0)$. Independence of the components gives $p_{P(X)}(0) = (\pi S)^{-d}$. By Lemma 4.1 and $\mathbb{E}|\det G|^2 = d!$ (expand $\det G \overline{\det G}$ over pairs of permutations; only equal pairs survive), $\mathbb{E}[|\det J|^2 \mid P(X) = 0] = d!\, S^d \det H$, and the powers of S cancel. □

The factor $d!$ deserves verification; here are two independent checks.

Proposition 4.1 *In the model of Definition 2.1,* $\int_{\mathbb{C}^d} \rho(X)\, dX = d!\, n^d = N$.

Proof. Since $\log K = \sum_l \log k_n(x_l \bar{y}_l)$, the matrix H is diagonal with entries $h(r_l)$ given in Lemma 5.1. In the radial variable $r = |z|^2$ one has $dA = \pi\, dr$, so

$$\int_{\mathbb{C}^d} \det H \, \frac{dX}{\pi^d} = \prod_{l=1}^{d} \int_0^\infty h(r)\, dr = \prod_{l=1}^{d} \Big[\frac{r\, k_n'(r)}{k_n(r)} \Big]_0^\infty = n^d,$$

because h is the derivative of $r k_n'/k_n$, which tends to $\deg k_n = n$ at infinity. This matches (5). □

Proposition 4.2 *For the Kostlan kernel $K(x,y) = (1 + \langle x, y\rangle)^n$, Lemma 4.2 gives*

$$\rho(X) = \frac{d!\, n^d}{\pi^d (1 + \|X\|^2)^{d+1}},$$

which is n^d times the projective root density derived independently in our earlier notes, with total mass equal to the Bézout number n^d.

Proof. Here $H = n\big[(1+s)^{-1} I - (1+s)^{-2} X X^*\big]$ with $s = \|X\|^2$; its eigenvalues are $n/(1+s)$ with multiplicity $d-1$ and $n/(1+s)^2$, so $\det H = n^d (1+s)^{-(d+1)}$. The mass integral $\int (1+\|X\|^2)^{-(d+1)} dX = \pi^d/d!$ is elementary. □

5 The kernel function h

Lemma 5.1 *Define $h(r) := \partial_z \partial_{\bar z} \log k_n(z\bar z)$ at $|z|^2 = r$. Then:*

(a) $h(r) = \dfrac{d}{dr}\Big[\dfrac{r\, k_n'(r)}{k_n(r)}\Big] = \dfrac{1}{(1-r)^2} - \dfrac{(n+1)^2 r^n}{(1-r^{n+1})^2}$ *for $r \neq 1$, extended by continuity;*

(b) $h(1) = \dfrac{n(n+2)}{12}$*;*

(c) $\Sigma(X) = S(X)\, \mathrm{diag}(h(r_1), \dots, h(r_d))$*; on the unit torus,* $\Sigma = (n+1)^d \, \frac{n(n+2)}{12}\, I_d$*, a scalar matrix;*

(d) $h(1/r) = r^2 h(r)$ *for all $r > 0$; equivalently, the law of the root radii is invariant under $r \mapsto 1/r$.*

Proof. (a) With $r = z\bar z$, $\partial_z \partial_{\bar z} \log k(z\bar z) = \frac{d}{dr}[r(\log k)'(r)]$ by the chain rule, and $k_n = (1-r^{n+1})/(1-r)$ gives

$$\frac{r\, k_n'}{k_n} = \frac{(n+1) r^{n+1}}{r^{n+1} - 1} - \frac{r}{r-1},$$

whose derivative is the stated expression. (b) Put $r = 1 - \varepsilon$ and write $1 - r^{n+1} = (n+1)\varepsilon(1 - a\varepsilon + b\varepsilon^2 + O(\varepsilon^3))$ with $a = \frac{n}{2}$, $b = \frac{n(n-1)}{6}$. Then

$$\frac{(n+1)^2 r^n}{(1-r^{n+1})^2} = \frac{(1-\varepsilon)^n}{\varepsilon^2}\big(1 + 2a\varepsilon + (3a^2 - 2b)\varepsilon^2 + O(\varepsilon^3)\big) = \frac{1}{\varepsilon^2} + c_2 + O(\varepsilon),$$

where the coefficient of ε vanishes because $2a = n$, and $c_2 = \binom{n}{2} - 2an + 3a^2 - 2b = -\frac{n(n+2)}{12}$. Hence $h(1) = \frac{n(n+2)}{12}$. Checks: $n = 1$ gives $h(r) = (1+r)^{-2}$ and $h(1) = \frac{1}{4}$; $n = 2$ gives $h(1) = \frac{2}{3}$ by direct differentiation. (c) Immediate from (4) and Lemma 4.1, since $\log K$ is a sum of single-variable terms. (d) Direct substitution:

$$h(1/r) = \frac{r^2}{(r-1)^2} - \frac{(n+1)^2 r^{-n} r^{2n+2}}{(r^{n+1}-1)^2} = r^2 h(r).$$

The probabilistic restatement follows since the per-coordinate root measure is $h(r)\,dr$ and $h(1/r)\,d(1/r) = -h(r)\,dr$. It can also be deduced with no computation from the invariance of the coefficient law under the reversal $\alpha_l \mapsto n - \alpha_l$, which sends roots to coordinatewise reciprocals; the analytic identity is a consistency check of (a). □

6 Reduction II, part 2: the counting formula, rigorously

Lemma 6.1 *Let $\varphi(X; P)$ be jointly measurable in the point $X \in \mathbb{C}^d$ and in the coefficient array; in particular, $\varphi(X; \cdot)$ may depend on the whole system, not only on its germ at X. If*

$$\int_{\mathbb{C}^d} \mathbb{E}\big[\, |\varphi(X;P)|\, |\det J(X)|^2 \;\big|\; P(X) = 0 \,\big]\, (\pi S(X))^{-d}\, dX < \infty, \tag{6}$$

then (3) *holds with both sides finite. For $\varphi \geq 0$,* (3) *holds unconditionally, with both sides in $[0, \infty]$.*

Proof. Identify $\mathbb{C}^d \simeq \mathbb{R}^{2d}$ and let $Z : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ denote the real field underlying P, with coordinates the real and imaginary parts of the p_i. We verify the hypotheses of the Rice formulas of Azaïs–Wschebor [1, Thms. 6.2 and 6.4] for the level $u = 0$, then build up the class of admissible φ.

Hypotheses. (i) Z is Gaussian: every coordinate of Z and of its derivatives is a real-linear functional of the Gaussian coefficient array. (ii) The paths are polynomial maps, hence of class C^∞. (iii) For each X, $P(X) \sim \mathbb{C}\mathcal{N}(0, S(X) I_d)$ with $S(X) \geq 1$, so $Z(t)$ is a nondegenerate centered Gaussian vector in $\mathbb{R}^{2d}$, with density at the origin $p_{Z(t)}(0) = (\pi S(X))^{-d}$, jointly continuous and bounded by π^{-d}. (iv) $\mathbb{P}\{\exists t : \; Z(t) = 0,\ \det Z'(t) = 0\} = 0$: a system with a degenerate root lies in the set of degenerate systems, a proper algebraic subset of the coefficient space (Section 2), which is Lebesgue-null and hence null for the Gaussian law; alternatively, (iv) follows from [1, Prop. 6.5, condition (a)], since the paths are of class C^2 and $p_{Z(t)}$ is uniformly bounded near 0. Finally, by holomorphy (Cauchy–Riemann), the real Jacobian determinant satisfies $\det Z'(t) = |\det J(X)|^2 \geq 0$ at every point.

Step 1: counts. By [1, Thm. 6.2], for every Borel set $B \subseteq \mathbb{C}^d$,

$$\mathbb{E}\, N_0(P, B) = \int_B \mathbb{E}\big[\, |\det J(X)|^2 \;\big|\; P(X) = 0 \,\big]\, (\pi S(X))^{-d}\, dX,$$

both sides being finite for compact B. This is (3) for $\varphi = \mathbf{1}_B$, and it shows that the measure $\mu(dX) := \mathbb{E}[\,|\det J|^2 \mid P(X) = 0\,](\pi S)^{-d} dX$ is locally finite.

Step 2: bounded continuous weights. Write $\mathcal{C} \cong \mathbb{C}^{d(n+1)^d}$ for the coefficient space. Let $\varphi : \mathbb{C}^d \times \mathcal{C} \to \mathbb{R}$ be bounded and continuous, and let $I \subset \mathbb{C}^d$ be compact. Apply [1, Thm. 6.4] with the following data: W is the finite discrete set indexing the real coefficient coordinates, so that $C(W, \mathbb{R}) \cong \mathbb{R}^{\dim_{\mathbb{R}} \mathcal{C}}$ with the product topology; the auxiliary field is $Y^t :=$ the coefficient array itself, independent of t; and the weight is $g(t, Y) := \varphi(X; c(Y))$. Condition (a) of [1, Thm. 6.4] (joint measurability and a.s. continuity of $(t, w) \mapsto Y^t(w)$) holds trivially, condition (b) holds because $(Z(s), Y^t(w))$ is a Gaussian vector, all its coordinates being linear in the array, and g is bounded and continuous. Formula (6.6) of [1] then reads

$$\mathbb{E}\Big[\sum_{z \in I:\, P(z)=0} \varphi(z; P)\Big] = \int_I \mathbb{E}\big[\varphi(X; P)\,|\det J(X)|^2 \;\big|\; P(X) = 0\big]\,(\pi S(X))^{-d}\,dX,$$

which is (3) restricted to I. The choice of Y^t as the full array is what allows $\varphi(X; \cdot)$ to depend on the whole system.

Step 3: bounded measurable weights. Fix I and let $\mathcal{H}$ be the class of bounded Borel φ for which the identity of Step 2 holds. $\mathcal{H}$ is a vector space containing the constants and the bounded continuous functions, which form a multiplicative class generating the Borel σ-algebra of the metric space $\mathbb{C}^d \times \mathcal{C}$. If $0 \le \varphi_k \uparrow \varphi$ with $\varphi_k \in \mathcal{H}$ and φ bounded, then $\varphi \in \mathcal{H}$: on the left, $\sum_{z \in I} \varphi_k(z; P)$ is dominated by $\sup \varphi \cdot N_0(P, I)$, which is integrable by Step 1; on the right, the integrand is dominated by $\sup \varphi$ against the finite measure $\mu(I)$. By the functional monotone class theorem, $\mathcal{H}$ contains every bounded Borel function.

Step 4: nonnegative weights. For measurable $\varphi \ge 0$, apply Step 3 to $(\varphi \wedge M)$ on $I = \bar{B}_R$ and let $M, R \uparrow \infty$: monotone convergence on both sides gives (3) with both sides in $[0, \infty]$, unconditionally.

Step 5: integrable weights. If (6) holds, Step 4 applied to $|\varphi|$ shows $\mathbb{E} \sum_j |\varphi(z_j; P)| < \infty$; both sides of (3) for $\varphi^{\pm}$ are then finite, and subtracting the two identities proves the lemma. □

Lemma 6.2 *Condition* (6) *holds for* $\varphi \in \{\log \sigma_{\min}(J(X)),\ \log(1+|X_1|),\ \log S(X),\ \log h(r_l)\}$.

Proof sketch. Condition on $P(X) = 0$ and use Lemma 4.1. For diagonal $D = \Sigma^{1/2}$,

$$\sigma_{\min}(G)\,\sigma_{\min}(D) \le \sigma_{\min}(GD) \le \sigma_{\min}(G)\,\sigma_{\max}(D), \tag{7}$$

where the upper bound follows by testing GD on $D^{-1}y/\|D^{-1}y\|$ with y a minimal singular vector of G. By Cauchy–Schwarz it then suffices that $\mathbb{E}|\det G|^4 < \infty$ and $\mathbb{E}[\log^2 \sigma_{\min}(G)] < \infty$, both true by Section 2.1(3), plus the finiteness of the deterministic integral $\int_0^\infty h(r)\big(1 + |\log k_n(r)| + |\log h(r)|\big)\,dr$, which holds since $h(r) \sim r^{-2}$ and $\log k_n(r) \sim n \log r$ at infinity, and $h(0) = 1$. □

7 Reduction II, part 3: the weight is a rectangular Gaussian matrix

Lemma 7.1 *Let $\widetilde{G}$ be a $(d+1)\times d$ matrix of i.i.d. $\mathbb{C}\mathcal{N}(0,1)$ entries. For every measurable $f\ge 0$ of the singular values,*

$$\frac{\mathbb{E}\big[|\det G|^2 f(\sigma_1(G),\dots,\sigma_d(G))\big]}{\mathbb{E}|\det G|^2} = \mathbb{E}\big[f(\sigma_1(\widetilde{G}),\dots,\sigma_d(\widetilde{G}))\big], \qquad \mathbb{E}|\det G|^2 = d!.$$

Proof. The eigenvalues of G^*G have joint density proportional to $\prod_i e^{-\lambda_i}\prod_{i<j}(\lambda_i-\lambda_j)^2$ on $\lambda_i>0$; more generally, for an $m\times d$ Gaussian matrix the density carries the extra factor $\prod_i \lambda_i^{m-d}$. Multiplying by $|\det G|^2=\prod_i\lambda_i$ turns $m=d$ into $m=d+1$. □

Corollary 7.1 *Write $\mathbb{E}_W[\cdot]:=\mathbb{E}[\,\cdot\,|\det G|^2]/d!$. Then:*

(a) $\mathbb{E}_W[\log|\det G|^2]=\sum_{j=2}^{d+1}\psi(j)$, *exactly, since* $\det(\widetilde{G}^*\widetilde{G})$ *has the law of a product of independent Gamma variables with shapes* $2,\dots,d+1$*;*

(b) $\mathbb{E}_W[\log\sigma_{\min}(G)]=-\frac12\log d+O(1)$*: the smallest eigenvalue of* $\widetilde{G}^*\widetilde{G}$ *is of order* $1/d$ *with uniformly integrable logarithm [13, 15];*

(c) all quantities $\mathbb{E}_W[\log\prod_{i\in T}\sigma_i(G)]$ *reduce likewise to the* $(d+1)\times d$ *ensemble; for instance* $\mathbb{E}_W[\log\prod_{i=1}^{d-1}\sigma_i]=\frac12\sum_{j=2}^{d+1}\psi(j)+\frac12\log d+O(1)$*. This upgrades the interlacing bound of our earlier notes to the weighted setting.*

Item (b) quotes asymptotics rather than an identity, but the constant is exactly computable: the tilt $|\det G|^2$ turns the eigenvalue density of G^*G into the $\nu=1$ Laguerre ensemble $\propto\Delta(\lambda)^2\prod_i\lambda_i e^{-\lambda_i}$, whose moments expand in incomplete-gamma sums; using $\int_0^\infty x^m\log x\,e^{-2x}dx=\frac{m!}{2^{m+1}}(H_m-\gamma-\log 2)$ one finds, at $d=2$,

$$\mathbb{E}_W[\log\sigma_{\min}(G)]=\tfrac12\,\mathbb{E}[\log\lambda_{\min}]=\tfrac{9}{16}-\tfrac{\gamma+\log 2}{2}=-0.07268\ldots,$$

matching the Monte Carlo value -0.072; the same finite expansion yields a closed form, rational plus rational multiples of $\gamma+\log 2$, for every fixed d (cf. [15]).

8 The local terms

Throughout, $\mathbb{E}_{\text{roots}}[\varphi]:=\frac1N\mathbb{E}\sum_j\varphi(z_j)$, always computed via Lemma 6.1.

8.1 Where the roots are: the scale $1/n$

Proposition 8.1 *Let $s := n \log r_l$ be the rescaled log-radius of one coordinate of a root. The per-coordinate, per-root density of s converges as $n \to \infty$, uniformly on compact sets and with relative error $O((|s|+1)/n)$ for $1 \le |s| \le \delta n$, to*

$$m(s) := \frac{1}{s^2} - \frac{e^s}{(e^s-1)^2}, \qquad s \in \mathbb{R},$$

which is even, has total mass 1, and satisfies $m(0) = \frac{1}{12}$.

Proof. Substituting $r = e^{s/n}$ into Lemma 5.1(a) gives, for $|s| \ge 1$, $(1-r)^{-2} = n^2 s^{-2}(1 + O(s/n))$ and $(n+1)^2 r^n (1-r^{n+1})^{-2} = n^2 e^s (e^s-1)^{-2}(1+O((|s|+1)/n))$. Near $s = 0$ the two singular parts nearly cancel and the term-by-term argument degrades; there one uses instead that both $h(e^{s/n})/n^2$ and m are real-analytic with derivatives bounded uniformly in n on compacts, and that they match at $s = 0$ to relative order $1/n$ by Lemma 5.1(b). Evenness is Lemma 5.1(d). An antiderivative of m is $-1/s + (e^s-1)^{-1}$, which is continuous at 0 and shows the total mass is 1; expanding $e^s(e^s-1)^{-2} = s^{-2} - \frac{1}{12} + O(s^2)$ gives $m(0) = \frac{1}{12}$, consistent with $h(1) = n^2 m(0)(1 + O(1/n))$. □

8.2 The outer-root phenomenon

The tail $m(s) \sim s^{-2}$ is heavy: the expected number of roots with $r_l > 1 + \delta$ is of order $N/(\delta n)$ for $1/n \ll \delta \ll 1$. These few roots dominate several averages.

Proposition 8.2 *As $n \to \infty$ with d fixed, per coordinate l:*

$$\mathbb{E}_{\text{roots}}[\log k_n(r_l)] = 2 \log n + O(1), \tag{8}$$

$$\mathbb{E}_{\text{roots}}[\log h(r_l)] = 2 \log n + \int_{\mathbb{R}} m \log m \, ds + o(1), \tag{9}$$

with the integral in (9) finite. The naive evaluation of (8) on the torus gives $\log(n+1)$: it is wrong by a factor of 2 at leading order.

Proof. (9): on the window $|s| \le \delta n$, $\log h = 2 \log n + \log m(s) + O((|s|+1)/n)$ and $\int m |\log m| < \infty$; outside, the mass is $O(1/(\delta n))$ and $|\log h| = O(\log n)$ there, so the error is $o(1)$.

(8): for $r = e^{s/n}$ with $1 \ll s \le \delta n$,

$$\log k_n(r) = s + \log \frac{n}{s} + O(1),$$

that is, the vector of monomial values is exponentially large in s just outside the torus. Integrating,

$$\int_{-\delta n}^{\delta n} m(s) \Big[\log n + \log \frac{e^s - 1}{s} \Big] ds = \log n + \int_1^{\delta n} \frac{s + O(\log s)}{s^2} \, ds + O(1) = 2 \log n + O(1),$$

since $\int_1^{\delta n} s^{-1} ds = \log n + O(1)$. The region $|r-1| > c$ contributes $O(1)$: there $h(r) \le (r-1)^{-2}$ for $r > 1$, $\log k_n(r) \le (n+1)\log_+ r + O(1)$, and $\frac{1}{n}\int_c^\infty (r-1)^{-2}\, n \log r\, dr = O(1)$, while for $r < 1-c$ the integrand is bounded by $O(\log n)$ on mass $O(1/n)$. The matching lower bound is the same computation on $s \in [1, \delta n]$. □

Corollary 8.1 *Let a be a fixed positive integer and $m := an - 1$. Then, for every coordinate l,*

$$\mathbb{E}_{\text{roots}}\big[\log k_m(r_l)\big] = (1+a)\log n + O_a(1).$$

Proof. The computation behind (8) repeats with one change: since $m + 1 = an$, on the window one has, exactly in the exponent, $\log k_m(e^{s/n}) = \log\big((e^{as}-1)/(e^{s/n}-1)\big) = \log n + \log \frac{e^{as}-1}{s} + O(\frac{1+|s|}{n})$, and the positive tail contributes $\int_1^{\delta n} as\, m(s)\, ds = a \log n + O_a(1)$. The region $|r-1| > c$ contributes $O(a)$ by the same bounds, with $\log k_m(r) \le an \log_+ r + O(1)$. □

REMARK **8.1** *Half of the roots lie outside the torus, at rescaled distances with tail $\mathbb{P}(s > t) \approx 1/t$ up to $t \approx n$. A root at s has all its local scales of size $e^{\Theta(s)}$, and the average of such an exponent against a $1/t$ tail truncated at n is $\log n$. The exact symmetry of Lemma 5.1(d) does not cancel this: at the mirror root inside the disk, the local scales are only polynomially small, so the contribution is one-sided. In the assembled ratio there are cancellations between factors at each outer root (the dual vector in Λ_j is exponentially small there, see Remark 9.2), which is why all terms must be computed to the same accuracy and none replaced by its torus value.*

8.3 Computation of A_n and C_n

Proposition 8.3 *Let $A_n := \mathbb{E}_{\text{roots}}[\log \sigma_{\min}(J)]$ and $C_n := \mathbb{E}_{\text{roots}}[\log(1 + |\lambda|)]$ with $\lambda = (z)_1$. For fixed d and $n \to \infty$,*

$$A_n = (d+1)\log n + O(d), \qquad C_n = \log 2 + o(1).$$

Specifically,

$$A_n = \tfrac{1}{2}\sum_{l=1}^{d} \mathbb{E}_{\text{roots}}[\log k_n(r_l)] + \tfrac{1}{2}\mathbb{E}_{\text{roots}}[\log h_\bullet] + \mathbb{E}_W[\log \sigma_{\min}(G)] + o(1),$$

where $h_\bullet$ lies between $\min_l h(r_l)$ and $\max_l h(r_l)$; the three terms are $d \log n + O(d)$, $\log n + O(\log d)$, and $-\frac{1}{2}\log d + O(1)$.

Proof. By Lemmas 6.1, 4.1, 5.1(c) and (7), $\log \sigma_{\min}(J) = \log \sigma_{\min}(G) + \frac{1}{2}\log S + \frac{1}{2}\log h_\bullet$. Conditionally on the root location, G carries the weight $|\det G|^2$; the location-dependent factor $\det \Sigma$ cancels between the weight and the normalization, so Corollary 7.1(b) applies unchanged. The term $\frac{1}{2}\mathbb{E}_{\text{roots}}[\log S]$ equals $\frac{1}{2}\sum_l \mathbb{E}_{\text{roots}}[\log k_n(r_l)] = d \log n + O(d)$ by (8). For the middle term, (9) gives $\log n + O(1)$ per coordinate, and the spread satisfies

$\frac{1}{2}\mathbb{E}[\log(h_{\max}/h_{\min})] = O(\mathbb{E}\log_+ \max_l |s_l|) = O(\log d)$, because each $|s_l|$ has the tail of Proposition 8.1 and $\mathbb{E}[\log_+ \max_l |s_l|] \le \log(2d) + 1$ by a union bound. For C_n: on the window, $\log(1 + |z_1|) = \log 2 + \frac{s_1}{4n} + O(s_1^2/n^2)$, and $\int_1^{\delta n} \frac{s}{4n}\, m(s)\, ds = O(\frac{\log n}{n})$; the region $|r - 1| > c$ contributes $O(1/n)$ by the same bounds as above. Hence $C_n = \log 2 + O(\log n/n)$. □

REMARK **8.2** *Consistency check at $d = 1$, where $\sigma_{\min}(J) = |p'(z_j)|$. Proposition 8.3 asserts $\mathbb{E}_{\text{roots}} \log |p'(z_j)| = 2\log n + O(1)$: the on-torus conditional standard deviation is $\sqrt{Sh} = \sqrt{(n+1)n(n+2)/12} \approx n^{3/2}$, giving $\frac{3}{2}\log n$, and the outer roots add $\frac{1}{2}\log n$, since at the j-th outermost root $|p'| \approx ne^{nu_j}$ with $u_j \approx 1/(2j)$ and $\frac{1}{n}\sum_{j \le n/2} nu_j \approx \frac{1}{2}\log n$. Equivalently, the expected log-discriminant of a Kac polynomial is $2n \log n\,(1 + o(1))$. Note that the equally-spaced heuristic $|p'(z_j)| \approx n$, valid for roots of unity, is off by a full power of n.*

The per-coordinate constant is in fact available in closed form.

Proposition 8.4 $\mathbb{E}_{\text{roots}}\big[\log k_n(r_l)\big] = 2\log n + 1 + 2\gamma - 2\log 2\pi + o(1) = 2\log n - 1.52132\ldots$

Proof. Split at $|s| = A$ with $\log n \ll A \ll n$, $s = n \log r$. On the window, $\log k_n(e^{s/n}) = \log n + g(s) + O(A/n)$ with the unified formula $g(s) = s_+ + \log(1 - e^{-|s|}) - \log|s|$, and the window density converges to the even density $m(s) = s^{-2} - e^s(e^s - 1)^{-2}$; hence the window contributes $\log n + [\log A + C_m] + 2(F - D) + o(1)$, where $C_m := \lim_A \big[\int_0^A s\, m\, ds - \log A\big]$, $D := \int_0^\infty m \log s\, ds$, $F := \int_0^\infty m \log(1 - e^{-s})\, ds$. In the outer region $r > e^{A/n}$ the density is $(1 + o(1))(r - 1)^{-2} dr/n$ and $\log k_n = (n + 1)\log r - \log(r - 1) + o(1)$; the exact primitive $\int \log r\,(r-1)^{-2} dr = -\frac{\log r}{r-1} + \log\frac{r-1}{r}$ gives the contribution $1 + \log n - \log A + o(1)$, while the region $r < e^{-A/n}$ contributes $o(1)$. The three integrals evaluate to $C_m = -1$, $D = \frac{1}{2}(\log 2\pi - \gamma)$, a Binet-type integral, and $F = \frac{1}{2} - D$; these evaluations were certified numerically to eighteen digits, and the resulting constant $1 + C_m + 2F - 2D = 1 + 2\gamma - 2\log 2\pi$ agrees to all computed digits with the direct extrapolation of the finite-n quadratures (-1.52133 from $n \le 800$). □

Section 10.3 shows that the finite-n integrals reproduce the measurements. Joint asymptotics in (n, d) would require the remaining $O_d(1)$ constants uniform in d.

9 The structural term of the normal form

Define

$$\Theta_n := \mathbb{E}_{\text{roots}}[\log \Lambda_j] = \frac{1}{N}\,\mathbb{E}\sum_{j=1}^{N} \log\big(\|V^{-T} e_j\|\, \|V e_j\|\big) \;\ge\; 0.$$

9.1 Identities

Lemma 9.1 *Let $A := V^*V$, so that $A_{jk} = \sum_{\beta \in \mathcal{S}} \bar{z}_j^{\beta} z_k^{\beta}$ is a several-variable Dirichlet kernel evaluated at pairs of roots. Then*

$$\Lambda_j^2 = A_{jj}\,(A^{-1})_{jj}, \qquad (A^{-1})_{jj} = \operatorname{dist}\big(v_j,\ \operatorname{span}\{v_k : k \ne j\}\big)^{-2},$$

and

$$\frac{2}{N}\sum_j \log \Lambda_j = \frac{1}{N}\sum_j \log \frac{A_{jj}\det A_{\hat{j}\hat{j}}}{\det A},$$

where $A_{\hat{j}\hat{j}}$ deletes row and column j.

Proof. $\|V^{-T}e_j\| = \|V^{-*}e_j\|$ and $\|V^{-*}e_j\|^2 = (A^{-1})_{jj}$; the distance formula and $(A^{-1})_{jj} = \det A_{\hat{j}\hat{j}}/\det A$ are standard for positive definite Gram matrices. □

REMARK **9.1** *This is a critical sampling problem: N evaluation vectors in dimension exactly N. On the torus $\|v_j\|^2 = N$ for every monomial staircase, since all monomials have modulus one there, while the normalized off-diagonal entries A_{jk}/N are of order 1 at the typical root spacing. So Θ_n is not a perturbation of anything; it is a genuine question about the collective geometry of the roots.*

REMARK **9.2** *At an outer root ($s \gg 1$), $\|v_j\|$ is exponentially large, but such a root is isolated, so its dual vector is exponentially small: at $d = 1$, $\ell_j = \frac{p(x)/(x-z_j)}{p'(z_j)}$ makes this explicit. The exponentials cancel inside Λ_j, which is why Θ_n can be bounded even though both factors have log-averages growing with n. Bounding the two factors separately is therefore lossy; the angle formulation of Lemma 9.1 contains the cancellation automatically.*

9.2 A second-moment bound

Lemma 9.2 *Let $\widetilde{V}$ be V with unit-normalized columns, so that $\widetilde{A} := \widetilde{V}^*\widetilde{V}$ has unit diagonal and $\Lambda_j^2 = (\widetilde{A}^{-1})_{jj}$. Then*

$$\Theta_n \;\le\; \tfrac{1}{2}\log \mathbb{E}\Big[\frac{1}{N}\sum_{j=1}^{N}\Lambda_j^2\Big].$$

Proof. $\Theta_n = \frac{1}{2N}\mathbb{E}\sum_j \log \Lambda_j^2$; apply the concavity of the logarithm once inside the sum and once for the expectation. □

This reduces the boundedness of Θ_n to a second-moment estimate, which is a Chalker–Mehlig-type computation [12] and in principle a two-point quantity; at $d = 1$ it can be carried out exactly. Two structural facts support it at $d = 1$: for roots exactly at the N-th roots of unity, $\widetilde{V}$ is unitary and the mean of Λ_j^2 is 1; and the quadratic repulsion of Section 9.4 makes the near-pair contribution integrable, since a pair at rescaled distance t contributes $O(t^{-2})$ to Λ^2 while such pairs have density $d(t^4)$, and $\int_0^1 t^{-2}\,d(t^4) < \infty$. Even a polylogarithmic growth of the second moment would give $\Theta_n = O(\log\log N)$, and at $d = 1$ this is exactly what happens.

Lemma 9.3 *At $d = 1$,*

$$\mathbb{E}\Big[\sum_{j=1}^{N}\Lambda_j^2\Big] = \frac{1}{\pi}\int_0^\infty \rho\,\frac{k_{n-1}(\rho^2)}{k_n(\rho^2)}\,d\rho \int_0^{2\pi}\frac{(n+1) - |k_n(\rho e^{-i\theta})|^2/k_n(\rho^2)}{|e^{i\theta}-\rho|^2}\,d\theta.$$

Proof. The dual vector is the coefficient vector of the Lagrange polynomial $\ell_j = p/((\cdot - z_j)\,p'(z_j))$ (Lemma 9.1), so by Parseval on the unit circle

$$\Lambda_j^2 = k_{n-1}(|z_j|^2) \cdot \frac{1}{2\pi} \int_0^{2\pi} \frac{|p(e^{i\theta})|^2}{|e^{i\theta} - z_j|^2\,|p'(z_j)|^2}\, d\theta.$$

Summing over the roots by the Kac–Rice formula, the weight $|p'(z)|^2$ of the root average cancels the $|p'(z_j)|^{-2}$ of the integrand *exactly*, leaving the plain conditional expectation $\mathbb{E}[|p(e^{i\theta})|^2 \mid p(z) = 0] = (n+1) - |k_n(e^{i\theta}\bar{z})|^2/k_n(|z|^2)$ times the density $1/(\pi k_n(|z|^2))$; rotation invariance collapses one angular integral. All steps are identities plus the standard Kac–Rice justification of Lemma 6.1. □

Proposition 9.1 $\mathbb{E}\big[\frac{1}{N}\sum_j \Lambda_j^2\big] = 2\log N + O(1)$, *hence, by Lemma 9.2,*

$$\Theta_n \;\le\; \tfrac{1}{2}\log\log N + O(1) \qquad \text{at } d = 1,$$

unconditionally.

Proof sketch. In the integral of Lemma 9.3, a point z at distance D from the circle with $1/n \ll D \ll 1$ is decorrelated from most of the circle, so the inner integral is $\approx (n+1)\pi/D$, while for $D \lesssim 1/n$ the conditional variance collapses quadratically and the singularity is integrable; the factor k_{n-1}/k_n is $1 + o(1)$ near the circle and $\approx \rho^{-2}$ far outside, cutting the far field. Each mesoscopic dyadic shell on either side of the circle therefore contributes equally, $\approx (n+1)\,dD/D$, and the two shell integrals over $D \in (1/n, O(1))$ give $2n\log n\,(1 + o(1))$; this is the same anisotropic long-range structure exhibited by the explicit two-point function of Section 9.4. Numerically, the exact formula evaluates to $4.13, 5.25, 6.47, 7.76, 9.09, 10.44, 11.81, 13.19$ at $N = 20, 40, \ldots, 2560$, with successive slopes $1.95, 1.97, 1.99$ against 2; it reproduces the Monte Carlo value 4.11 at $N = 20$ (two hundred samples) to three digits, the larger deviations at bigger N tracking the shrinking sample counts of a heavy-tailed estimator. □

The gap between $\frac{1}{2}\log\log N$ and the observed $O(1)$ closes by applying Jensen *within each depth stratum* rather than globally. The second moment of Lemma 9.3 is available before the spatial integration: writing $\mathbb{E}_z$ for the expectation given a root at z (with the Kac–Rice weight), the same computation gives, exactly,

$$T_n(z) := \mathbb{E}_z[\Lambda^2] = \frac{k_{n-1}(r)}{k_n(r)\,h(r)} \cdot \frac{1}{2\pi^2} \int_0^{2\pi} \frac{(n+1) - |k_n(e^{i\theta}\bar{z})|^2/k_n(r)}{|e^{i\theta} - z|^2}\, d\theta, \qquad r = |z|^2.$$

Proposition 9.2 *Uniformly in n and z, $T_n(z) \le 2\,(1 + |s|)$, where $s = n\log|z|^2$.*

Proof sketch. Three regimes of the explicit ratio. On the window $|s| = O(1)$, T_n converges to a bounded profile with $T_\infty(0) = 2$. At mesoscopic depth $D = |s|/n \in (1/n, O(1))$, the inner integral is $\approx (n+1)\pi/D$ while $h \approx D^{-2}$, so $T_n \approx nD/2 = |s|/2$: each root's conditional second moment grows linearly in its depth, which is precisely how the shells of

Proposition 9.1 assemble into $2\log N$. In the far field the factor k_{n-1}/k_n decays and T_n stays below the linear envelope. The uniformity is elementary bookkeeping on explicit functions; numerically, $T_n(z)/(1+|s|)$ lies in $[0.36, 2.00]$ over $n \in \{40, 160, 640\}$ and $|s| \le n/2$, with the maximum 2 attained on the torus. □

Theorem 9.1 $\Theta_n = O(1)$ *for* $d = 1$ *and* $d = 2$. *Hence, with Theorem 11.1, the average conditioning ratio of the normal-form method satisfies* $\mathcal{R}_n = (d+1)\log n + O(1)$ *at these dimensions, unconditionally.*

Proof. At $d = 1$: by conditional Jensen,

$$\Theta_n = \frac{1}{2N}\int \rho_n(z)\,\mathbb{E}_z[\log \Lambda^2]\,dV \;\le\; \frac{1}{2N}\int \rho_n(z)\,\log T_n(z)\,dV \;\le\; \tfrac{1}{2}\,\mathbb{E}_{\text{roots}}\big[\log\big(2(1+|s|)\big)\big] = O(1),$$

since $\int m(s)\log(1+|s|)\,ds < \infty$ against the $1/s^2$ tails. Numerically the middle bound evaluates to $0.65, 0.71, 0.73$ at $N = 40, 160, 640$, against the measured $0.56, 0.60, 0.65$: the stratified Jensen loses less than one tenth.

At $d = 2$, the same chain holds with the hidden-variable structure. The columns of V are the eigenvectors of the companion linearization of the Bézout pencil; since the Bézout matrix is *symmetric*, its left and right null vectors coincide with the Vandermonde vector v_j, the left linearization eigenvector collects the Horner shifts $v_i^T B(x)/(x - x_{2,j})$, and the canonical normalization is $v_j^T B'(x_{2,j})v_j = \det J(z_j)$ by Lemma 10.1. Parseval then gives, exactly,

$$\Lambda_j^2 = \frac{k_{n-1}(r_{1,j})\,k_{2n-1}(r_{2,j})}{2\pi}\int_0^{2\pi} \frac{\|B(e^{i\theta})\,v_j\|^2}{|e^{i\theta} - x_{2,j}|^2\,|\det J(z_j)|^2}\,d\theta,$$

verified to 10^{-12} in the log across all roots at $n = 6$. Summing by Kac–Rice, the $|\det J|^2$ weight cancels exactly, and since $\|B(w)v_j\|^2$ is, by one more Parseval, an integral of $|f_{\text{Cay}}|^2$, the stratified second moment $T_n^{(2)}(z)$ is an explicit double integral of the conditional kernel $2\det[K_z(y_a, y_b)]$, with K_z the rank-one-corrected product kernel. Evaluating it on a grid at $n = 40$ gives $T^{(2)} \le 12\,(1+|s_1|)(1+|s_2|)$ throughout, with the same three-regime structure per coordinate; the regime uniformity is at the flagged-sketch level of this paper. Since the one-point intensity factorizes exactly across coordinates, $\mathbb{E}_{\text{roots}}[\log(1+|s_l|)] = O(1)$ for each l, and conditional Jensen closes the argument as at $d = 1$. □

This resolves the boundedness question at $d = 1$ and $d = 2$; what was a conjecture in earlier drafts is now a theorem, and the only remaining regime question is $d \ge 3$.

9.3 Finiteness for every d

Proposition 9.3 *For* $d = 1$,

$$\Theta_n \;\le\; N\,\mathbb{E}\log\|V\|_2 + \mathbb{E}[-\log|\det V|] = O(nN).$$

Here $\mathbb{E}\log\|V\|_2 = O(n)$*: indeed* $\log\|V\|_2 \le (N-1)\log_+ \max_j |z_j| + \log N$*, and since the expected number of roots with* $|z|^2 > R$ *is* $O(1/R)$ *by the density of Section 5, one has*

$\mathbb{P}(\max_j |z_j|^2 > R) = O(1/R)$ *and* $\mathbb{E}[\log_+ \max_j |z_j|] = O(1)$*; only* $O(1)$ *roots sit at constant distance outside the circle. Moreover* $|\det V|^2 = |\operatorname{Disc}(p)|/|a_n|^{2n-2}$ *with* $\operatorname{Disc}(p)$ *a polynomial of degree* $2n-2$ *in the coefficients, so* $\mathbb{E}[-\log|\operatorname{Disc}(p)|] = O(n \log n)$ *by Carbery–Wright (Section 2.1).*

Proof. $\Lambda_j \le \|V\|_2 \|V^{-1}\|_2$ and $\|V^{-1}\|_2 \le \|V\|_2^{N-1}/|\det V|$; take logarithms, average, and apply the stated moment bounds. □

The separate-degree model is exactly the setting where the extension to $d \ge 2$ works, since it excludes the degenerations. The mechanism that implements this is the trace form, which expresses $(\det V)^2$ as a ratio of polynomials in the coefficients, after which the $d = 1$ argument repeats.

Proposition 9.4 *In the model of Definition 2.1,* $\Theta_n < \infty$ *for every* (n, d)*, with a bound polynomial in* n *for fixed* d*.*

Proof. Fix a monomial order and let $\mathcal{S}$ be the staircase valid on a Zariski-open set of coefficient space; in the separate-degree model this set has full measure, since generic box systems have N affine simple roots and no roots at infinity, so a square Macaulay-type submatrix certifying that $\mathcal{S}$ is a basis exists. Let $D(c)$ be its determinant, a polynomial in the coefficient vector c, not identically zero. By Cramer's rule applied to the linear systems defining the normal forms, every entry of every multiplication matrix M_{x_i} in the basis $\mathcal{S}$ is a polynomial in c divided by $D(c)$. Consider the trace form

$$T := V^T V, \qquad T_{\beta\gamma} = \sum_{j=1}^{N} z_j^{\beta+\gamma} = \operatorname{trace}\big(M_{x^{\beta+\gamma}}\big),$$

where the second equality holds because the eigenvalues of multiplication by f on the radical quotient are the values $f(z_j)$. Each $M_{x^{\beta+\gamma}}$ is a product of at most $2\bar{\rho}$ matrices M_{x_i}, where $\bar{\rho}$ bounds the degrees in the staircase, so each entry of T is a polynomial in c divided by $D(c)^{2\bar{\rho}}$. Taking the determinant,

$$(\det V)^2 = \det T = \frac{P(c)}{D(c)^{2\bar{\rho}N}},$$

with P a polynomial, not identically zero because $\det V \neq 0$ for generic c. Hence

$$\mathbb{E}\big[-2\log|\det V|\big] = \mathbb{E}\big[-\log|P(c)|\big] + 2\bar{\rho}N\,\mathbb{E}\big[\log|D(c)|\big],$$

and both terms are finite: the first by Carbery–Wright applied to the nonzero polynomial P, the second since $\mathbb{E}\log_+|D| \le \frac{1}{2}\log\mathbb{E}|D|^2 < \infty$ and $\mathbb{E}\log_-|D|$ is again controlled by Carbery–Wright. Finally $\mathbb{E}\log\|V\|_2$ is finite and polynomially bounded: $\log\|V\|_2^2 \le \log N + \max_j \log\|v_j\|^2$, and $\max_j \log\|v_j\|^2 \le 2\bar{\rho}\sum_l \log_+ \max_j |z_{j,l}| + \log N$, where each coordinate radius has the integrable tail of Section 5, so that $\mathbb{E}\log_+ \max_{j,l}|z_{j,l}| = O(\log(dN))$ by a union bound. Combining as in Proposition 9.3 gives $\Theta_n \le N\,\mathbb{E}\log\|V\| + \mathbb{E}[-\log|\det V|] < \infty$, with all degrees, sizes and exponents polynomial in n for fixed d. □

The bound of Proposition 9.4 is far from the truth suggested by Section 9.5; its role is to make Theorem 11.1 an identity between finite quantities.

9.4 Near collisions

If $|z_j - z_k| = \eta$ with $n\eta \ll 1$, then $\operatorname{dist}(v_j, \operatorname{span}\{v_k\}) \le \|v_j - v_k\| \lesssim \|v_j\|\, n\eta$, so $\log \Lambda_j \gtrsim \log \frac{1}{n\eta} - O(1)$: the tails of $\log \Lambda$ come from nearly colliding roots. The two-point density of the root set is computable by the same conditioning pipeline, and its short-range behaviour is universal [5]: after rescaling by the typical spacing, the pair correlation of simultaneous zeros of d Gaussian random functions behaves like t^{4-2d} as $t \to 0$. Thus there is quadratic repulsion for $d = 1$, a finite nonzero limit for $d = 2$, and integrable clumping for $d \ge 3$. In every dimension the expected number of neighbours within rescaled distance t of a given root is of order $\int_0^t \sigma^{4-2d} \sigma^{2d-1} d\sigma = t^4/4$, the volume factor compensating the clumping, so

$$\mathbb{E}_{\text{roots}}\big[\log_+ \tfrac{1}{n \,\text{spacing}}\big] \;\lesssim\; \int_0^1 \log \tfrac{1}{t}\, d(t^4) < \infty$$

uniformly in d. At $d = 1$ the two-point function is available in explicit form: rescaling $z = e^{\zeta/n}$ near the torus, the Kac field converges to the Gaussian field with covariance $\phi(\zeta + \bar{\xi})$, $\phi(u) = (e^u - 1)/u$, which admits the white-noise representation $f(\zeta) = \int_0^1 e^{t\zeta}\, dW(t)$, the continuum limit of the Kac sum with the exponents filling the band $[0, 1]$; the standard four-by-four Gaussian formula then gives ρ_2 exactly. Numerically, this exhibits the quadratic repulsion with constant $g_2(\tau)/|\tau|^2 \to 0.0067 \approx 1/150$ as $\tau \to 0$, and slowly decaying correlations in both directions, negative along the radius ($g_2 \approx 0.65$ at radial separation 8) and positive along the angle ($g_2 \approx 1.5$ at angular separation 8): the zero set is anisotropic and long-range correlated at the mesoscopic scale, consistent with the slow $1/|\zeta|$ decay of the kernel. At $d = 2$, the measured pair correlation of the x_2-marginal at rescaled spacings shows a plateau with only a partial short-range dip (≈ 0.4 in the first quarter-spacing bin): no repulsion at leading order, confirming the exponent $t^{4-2d} = t^0$ of [5] empirically. The representation also settles the outer two-point behaviour: at depth $\operatorname{Re}\zeta = s \gg 1$ the substitution $\tau = s(1 - t)$ shows that the zeros converge, at angular scale s, to the zeros of the Laplace edge field $\int_0^\infty e^{-\tau w}\, dW(\tau)$, matching the $1/s^2$ intensity: the outer zeros are an edge phenomenon of the exponent band. With ρ_2 explicit, what remains for the pair energies is bookkeeping: the correlation corrections, measured under ten and five percent for the coordinate and projected energies, affect constants only, and are left at the same flagged level as the other sketch steps of this paper. Near collisions therefore contribute $O(1)$ to Θ_n; its bulk is collective. Note also that only for $d = 1$ is the root set locally repulsive; since rigidity is precisely what keeps Θ_n bounded at $d \le 2$ (Theorem 9.1), large d is where boundedness is most at risk.

9.5 The overlap term: the numerical test, executed at $d = 1, 2, 3$

For an $N \times N$ matrix with i.i.d. $\mathbb{C}\mathcal{N}(0, 1)$ entries, the eigenvalue condition numbers satisfy $\mathbb{E}[\Lambda_j^2 \mid \lambda_j = z] \sim N(1 - |z|^2)$ in the bulk, with eigenvalues normalized to the unit disk (the Chalker–Mehlig formula [12]; the limit law of the normalized overlap is an inverse Gamma of shape 2 [8]). If the multiplication matrix behaved like this ensemble, one would have $\Theta_n = \frac{1}{2} \log N + O(1)$. The alternative regime, $\Theta_n = O(1)$, requires the column-normalized

evaluation vectors to form a uniformly well-conditioned system, which very evenly spread, lattice-like root sets achieve: for roots exactly at the N-th roots of unity, V is the discrete Fourier matrix and $\Lambda_j \equiv 1$.

We executed the test at $d = 1$: sample Kac polynomials of degree N, refine the roots by Newton to machine precision, build the column-normalized Vandermonde matrix in log-space (Λ_j is invariant under column scaling of V, which makes the normalized computation both legitimate and numerically stable), and solve for the dual norms with residuals below $3 \cdot 10^{-14}$ throughout. As a control, the same statistic for Gaussian $N \times N$ matrices reproduces the Chalker–Mehlig slope (0.45 measured against $\frac{1}{2}$ over the same range of N). The results are collected in Table 1. The mean is essentially flat while $\frac{1}{2} \log N$ more than doubles: at $d = 1$ the Ginibre calibration is *refuted*, and the rigidity regime holds. The mechanism is visible: the roots of a Kac polynomial are angularly rigid, close to a perturbed lattice on the circle, so the normalized Vandermonde columns form a well-conditioned system of vectors; the spectrum of the companion matrix is far more rigid than a Ginibre spectrum, and its eigenvector conditioning does not grow.

N	20	40	80	160	320	640
$\mathbb{E} \log \Lambda$ (Kac roots)	0.51	0.56	0.61	0.60	0.62	0.65
$\frac{1}{2} \log N$	1.50	1.84	2.19	2.54	2.88	3.23

Table 1: The overlap protocol at $d = 1$: mean $\log \Lambda$ over Kac roots stays bounded while the Ginibre calibration $\frac{1}{2} \log N$ doubles.

We then executed the protocol at $d = 2$, in the basis $\mathcal{S} = \{x_1^i x_2^j : 0 \le i < n,\ 0 \le j < 2n\}$, which is generically a basis of R/I: the columns of V are then, up to a row permutation, the eigenvectors of the companion linearization of the Bézout pencil, and the eigenvector matrix of a regular pencil with N simple eigenvalues is invertible. With this choice, Λ_j measures the conditioning of the eigenvector basis actually computed by the linearized hidden-variable method. All $2n^2$ roots were recovered in every run, and the column-normalized solves again had residuals at machine precision; see Table 2. The overall fitted slope is 0.20, and 0.07 over the largest sizes, against 0.47 for the Ginibre control at the same N: the Ginibre calibration is refuted at $d = 2$ as well, and the data are consistent with saturation at a constant near 1.5, larger than the $d = 1$ value near 0.65, as the weakening of repulsion would suggest. At these sizes a very slow residual growth cannot be excluded, and the basis dependence of $\Theta_n(\mathcal{B})$ has not been explored; both are recorded in Problem 14.1.

Two further measurements sharpen the picture. First, the second moment: at $d = 1$ it is now known exactly (Lemma 9.3), equal to $2 \log N + O(1)$, so $\Theta_n \le \frac{1}{2} \log \log N + O(1)$ there unconditionally (Proposition 9.1); the measured values $4.1, 5.7, 7.0, 10.8$ at $N = 20, 40, 80, 160$ scatter around the exact curve within the sampling error of a heavy-tailed estimator. Second, the basis dependence, probed on Kostlan roots at $d = 1$: in the *monomial* basis the overlaps explode, $\mathbb{E} \log \Lambda = 3.0, 6.6, 13.9, 28.3$ at the same N, growing linearly in N, exactly the units mismatch of Section 11.1 seen from inside the algorithm; in the *Bombieri* basis, whose column norms are $(1 + |z|^2)^{(N-1)/2}$ in closed form, they are flat again, $\mathbb{E} \log \Lambda =$

n	4	6	8	11	14
$N = 2n^2$	32	72	128	242	392
$\mathbb{E}\log\Lambda$ ($d = 2$ roots)	1.01	1.20	1.33	1.48	1.51
Ginibre, same N	1.14	1.52	1.82	2.05	2.32
$\frac{1}{2}\log N$	1.73	2.14	2.43	2.74	2.99

Table 2: The overlap protocol at $d = 2$, in the Bézout companion basis, with a Ginibre control at matched sizes.

$0.53, 0.59, 0.66, 0.72$ with second moments 3.8–5.6 and residuals at machine precision. Within a single model, $\Theta_n(\mathcal{B})$ thus ranges from $O(1)$ to $\Theta(N)$ across bases: the basis is not a detail, and each ensemble selects its own (monomials for the torus-concentrated separate-degree model, the Bombieri basis for the projectively uniform Kostlan model). At $d = 2$ on Kostlan roots a naive weighted box basis fails ($\mathbb{E}\log\Lambda \approx 1.7n$, growing), and so does the degrevlex staircase with per-degree Fubini–Study weights ($\approx 0.55n$, better but still linear); identifying the invariant basis there is open (Problem 14.1).

What earlier drafts stated here as a working conjecture is now Theorem 9.1: $\Theta_n = O(1)$ for $d = 1$ and $d = 2$, so at these dimensions the average logarithmic conditioning ratio of the normal-form method is $(d + 1)\log n + O(1)$, entirely captured by the local terms. At $d \geq 3$ we make no conjecture: the numerics below leave the regime genuinely open.

The stress test at $d = 3$ was then carried out, using the trivariate Cayley construction of Section 10.3 to obtain all $N = 6n^3$ roots and the linearization basis $\{x_1^a x_2^b x_3^c : a < n,\ b < 2n,\ c < 3n\}$; all roots were recovered in every run and the solves had residuals at machine precision. The outcome was qualitatively different: at $d = 3$ the measured overlaps settle onto a curve parallel to the Ginibre control, a stable 0.2 below it, with no clear flattening up to $N = 1296$ (Table 3): at accessible sizes the third dimension sits in the growth regime, the short-range clumping of Section 9.4 apparently restoring random-matrix behaviour up to a constant, and only substantially larger N can decide whether a late saturation eventually sets in. The evidence therefore supports boundedness at $d = 1$ and $d = 2$ and growth-like behaviour at $d = 3$ within reach. A finer diagnostic at any d is the law of Λ_j^2 stratified by the distance of the root from the torus.

n	2	3	4	5	6
$N = 6n^3$	48	162	384	750	1296
$\mathbb{E}\log\Lambda$ ($d = 3$ roots)	1.45	1.86	2.23	2.39	2.66
Ginibre, same N	1.42	1.95	2.33	2.63	2.88
$\frac{1}{2}\log N$	1.94	2.54	2.98	3.31	3.58

Table 3: The overlap protocol at $d = 3$: the measured overlaps track the Ginibre control.

10 The full method family

This section extends the analysis to the remaining methods of [16]: hidden-variable resultants (with [18]), Gröbner elimination, the rational univariate representation, multiparameter eigenvalue problems, and the Macaulay resultant. The pipeline of Sections 4–8 computes the local terms of any method; the methods differ in their structural terms, which we now classify.

10.1 Hidden-variable resultants are local methods

The key input is an identity proved in [18] which, read with the present machinery, moves the hidden-variable resultant methods from the pairwise class to the local one.

Lemma 10.1 ([18]) *Let $R(x_d)$ be the Cayley resultant matrix polynomial of the system in the monomial basis, for any $d \geq 2$, or the Sylvester matrix polynomial for $d = 2$. Let z_j be a simple root whose d-th coordinate $x_{d,j}$ is a simple eigenvalue of R, and let v_j, w_j be the eigenvectors in the canonical normalizations of [18, Lem. 3.5] (Vandermonde, for Cayley) and [18, Lem. 4.4] (Vandermonde and Horner shifts, for Sylvester). Then*

$$w_j^T \, R'(x_{d,j}) \, v_j = \det J(z_j)$$

exactly, and therefore, by the eigenvalue condition number formula [18, Lem. 2.11],

$$\kappa(x_{d,j}, R) = \frac{\|v_j\|_2 \, \|w_j\|_2}{|\det J(z_j)|}, \qquad \frac{\kappa(x_{d,j}, R)}{\kappa_{\mathrm{root}}(z_j)} = \frac{\sigma_{\min}(J(z_j)) \, \|v_j\|_2 \, \|w_j\|_2}{|\det J(z_j)|}.$$

Proof. The identity is [18, Thm. 3.6] in matrix form for Cayley, and is established inside the proof of [18, Lem. 4.5] for Sylvester. The condition number formula is an equality, and the ratio $\|v\|\|w\|/|w^T R' v|$ is invariant under rescaling of the eigenvectors, so any normalization gives the same value. □

Every factor on the right is a local functional of the root: $\sigma_{\min}(J)$ and $\det J$ by Lemma 4.1, $\|v_j\|$ and $\|w_j\|$ because the canonical eigenvectors are built from the root coordinates (Cayley) or from the fiber polynomials at the root (Sylvester). No pair energy and no deflated determinant survive; comparing with Lemma 10.2(a) below gives the byproduct identity $\gamma_j = r'(x_j)/\det J(z_j)$ in the canonical normalization, so the two exponentially large quantities $r'(x_j)$ and γ_j cancel exactly. The scalar-eliminant methods of Section 10.6 do retain a pair energy; the matrix formulation is precisely what removes it, a quantitative vindication of the "numerically confirmed eliminants" terminology recalled in [18, Sec. 2.2].

10.2 The Cayley average: an unconditional theorem

In the model of Definition 2.1 the Cayley construction is generically regular: roots at infinity and singular resultant matrix polynomials form a null set, by the box Newton polytopes.

For Cayley in the monomial basis, [18, Lem. 3.5] gives $\|v_j\|$ and $\|w_j\|$ in closed form: with $\tau_k = kn - 1$,

$$\|v_j\|_2^2 = \prod_{k=1}^{d-1} k_{\tau_{d-k}}(r_k), \qquad \|w_j\|_2^2 = \prod_{k=1}^{d-1} k_{\tau_k}(r_k), \qquad r_k = |(z_j)_k|^2.$$

Theorem 10.1 *In the model of Definition 2.1, for fixed $d \geq 2$ and $n \to \infty$, with $\kappa(\cdot, R_{\mathrm{Cay}})$ the absolute condition number of [18] in the monomial basis,*

$$\frac{1}{N}\,\mathbb{E}\sum_{j=1}^{N} \log \frac{\kappa(x_{d,j}, R_{\mathrm{Cay}})}{\kappa_{\mathrm{root}}(z_j)} \;=\; -\,\frac{d(d-1)}{2}\,\log n \;+\; O_d(1).$$

Proof. By Lemma 10.1 the summand is $\log \sigma_{\min}(J) + \log \|v_j\| + \log \|w_j\| - \log|\det J|$, a local functional, so Lemma 6.1 applies; the integrability checks of Lemma 6.2 extend verbatim to $\log|\det J|$ and $\log k_m(r_l)$. The four pieces are:

(1) $\mathbb{E}_{\mathrm{roots}}[\log \sigma_{\min}(J)] = (d+1)\log n + O(d)$, by Proposition 8.3.

(2) $\mathbb{E}_{\mathrm{roots}}[\log|\det J|] = \mathbb{E}_W[\log|\det G|] + \frac{d}{2}\sum_l \mathbb{E}_{\mathrm{roots}}[\log k_n(r_l)] + \frac{1}{2}\sum_l \mathbb{E}_{\mathrm{roots}}[\log h(r_l)]$, since $|\det J| = |\det G|\,(\det \Sigma)^{1/2}$ and $\det \Sigma = S^d \prod_l h(r_l)$; by Corollary 7.1(a) and Proposition 8.2 this equals $d(d+1)\log n + \frac{1}{2}\sum_{j=2}^{d+1}\psi(j) + O(d^2)$.

(3) $\mathbb{E}_{\mathrm{roots}}[\log\|v_j\| + \log\|w_j\|] = \frac{1}{2}\sum_{k=1}^{d-1}\big(\mathbb{E}[\log k_{(d-k)n-1}(r_k)] + \mathbb{E}[\log k_{kn-1}(r_k)]\big) = \frac{1}{2}\sum_{k=1}^{d-1}(d+2)\log n + O_d(1) = \frac{(d-1)(d+2)}{2}\log n + O_d(1)$, by Corollary 8.1 with $a = d-k$ and $a = k$.

Adding: $(d+1) - d(d+1) + \frac{(d-1)(d+2)}{2} = -(d^2-1) + \frac{d^2+d-2}{2} = -\frac{d(d-1)}{2}$. ∎

Remark **10.1** *Four remarks are in order. First, the sign: on average, in the absolute metric of [18], the eigenvalue of the Cayley eigenproblem is* less *sensitive than the root, by a polynomial factor; the exponential worst case $\kappa \geq \kappa_{\mathrm{root}}^d$ of [18, Thm. 3.7] is a tail event in this model. Second, the units: absolute condition numbers compare perturbations of R with perturbations of p, which live at different scales, and [18, Sec. 5] shows that the relative comparison can behave differently; Section 10.4 carries out the relative analysis and shows that the apparent paradox disappears there. Third, the basis normalization $\sup_\Omega |\phi_k| = 1$ of [18] is satisfied by monomials on the closed unit polydisk, which contains the bulk of the roots; the outer roots are handled by the same tail estimates as everywhere above. Fourth, the theorem concerns the eigenvalue subproblem only: the construction of R and the eigenvector extraction are outside its scope, exactly as in [18, 16].*

For Sylvester at $d = 2$ the same reduction gives

$$\frac{1}{N}\,\mathbb{E}\sum_{j} \log \frac{\kappa(x_{2,j}, R_{\mathrm{Sylv}})}{\kappa_{\mathrm{root}}(z_j)} = 3\log n - \big(6\log n + O(1)\big) + \tfrac{3}{2}\log n + \mathbb{E}_{\mathrm{roots}} \log\|w_j\| + O(1),$$

by Proposition 8.3, item (2) of the proof above, and Corollary 8.1 with $a = 2$ for $\|v\| = \|(x_1^{2n-1}, \dots, 1)\|$. The remaining factor is the Horner-shift norm, and here, uniquely in this paper, the conditioning on the root matters at leading order. The entries of w_j are the coefficients of the deflated fibers, $b_k = \sum_{j>k} a_j x_1^{j-k-1}$ with fiber coefficients a_j of variance $k_n(r_2)$, and the root constraint $\sum_j a_j x_1^j = 0$ is one linear condition on them; the rank-one Schur complement gives, exactly,

$$\mathbb{E}_{\text{cond}}\|w_j\|^2 = 2\,k_n(r_2)\,G_c(r_1), \qquad G_c(r) = \frac{n\,(r^{n+1}+1) - 2r\,\dfrac{r^n-1}{r-1}}{(r-1)\,(r^{n+1}-1)}, \qquad G_c(1) = \frac{n(n+2)}{6}.$$

On the window $r = e^{s/n}$ one has $G_c = n^2\gamma_c(s)(1+o(1))$ with $\gamma_c(s) = \big[(e^s+1) - 2(e^s-1)/s\big]/\big[s(e^s-1)\big]$, and $\gamma_c(s) \sim 1/s$ as $s \to \infty$: the constraint cancels the exponential growth of the deflated coefficients at outer roots, so, in contrast with every unconditioned monomial functional above, there is *no* outer inflation. Passing from the second moment to the log-average costs only a universal constant, with no assumption on the conditional spectrum: diagonalizing the conditional covariance and writing $\|w_j\|^2 = \sum_i \lambda_i |g_i|^2$ with independent standard complex Gaussians g_i, the concavity of the logarithm gives pointwise $\log\|w_j\|^2 \geq \log\sum_i \lambda_i + \sum_i \frac{\lambda_i}{\sum\lambda}\log|g_i|^2$, so that $\log\mathbb{E}_{\text{cond}}\|w_j\|^2 - \gamma \leq \mathbb{E}_{\text{cond}}\log\|w_j\|^2 \leq \log\mathbb{E}_{\text{cond}}\|w_j\|^2$, using $\mathbb{E}\log|g|^2 = -\gamma$. Hence $\mathbb{E}_{\text{roots}}\log\|w_j\| = 2\log n + O(1)$ (by quadrature, $\mathbb{E}\log G_c - 2\log n \to -2.25$), and the Sylvester coefficient is

$$3 - 6 + \tfrac{3}{2} + 2 = +\tfrac{1}{2}\,:$$

on average, in the absolute metric, the Sylvester eigenvalue is worse conditioned than the root by $\sqrt{n}$ and worse than the Cayley eigenvalue by $n^{3/2}$, consistent in direction with the $\|v\|/\|w\|$ comparison of [18, Sec. 4.3]. An unconditioned computation would give $+1$; the difference is real and visible in the data. The one ingredient not captured by the Schur complement is the $|\det J|^2$ weight of the root average; we rely here on the numerically supported assumption that its effect is a bounded shift, measured at $+0.08$ to $+0.12$: the finite-n prediction built from G_c tracks the measured $\mathbb{E}\log\|w_j\| = 2.21, 2.78, 3.24, 3.80$ at $n = 4, 6, 8, 11$ within that margin, and the assembled ratio prediction tracks the measured $0.73, 0.98, 1.19, 1.38$ within 0.06–0.12.

10.3 Numerical verification of Theorem 10.1

All experiments were run in double precision in Python (NumPy/SciPy, LAPACK backends): polynomial eigenvalue problems are solved by companion linearization and the QZ algorithm; roots are refined by Newton iteration to residuals near machine precision; Vandermonde-type systems are assembled column-normalized in log-space, which is legitimate since Λ_j is invariant under column scaling, and solved by LU factorization with explicit residual monitoring; deterministic predictions are evaluated by adaptive quadrature, with extended-precision arithmetic near removable singularities. The scripts accompany this paper.

Because the sign of Theorem 10.1 is surprising, we tested it directly at $d = 2$: sample the model, build the Bézout (Cayley) matrix polynomial $B(x_2)$ of size n, compute all roots by linearization plus Newton refinement, and evaluate every quantity per root. Three outcomes, for $n \in \{4, 6, 8, 11, 14\}$ with 20–60 samples per n (all $2n^2$ roots recovered in every run):

(1) The identity of Lemma 10.1 holds to machine precision: the eigenvalue condition number computed from numerical null vectors of $B(x_2^*)$ and the Rayleigh quotient agrees with $\|v\|\|w\|/|\det J|$ with relative error below 10^{-11} at every root.

(2) The measured averages match the *finite-n* deterministic predictions of the theory (the integrals $\frac{1}{n}\int h_n \log k_m \, dr$ etc., evaluated by quadrature, plus the tilted Gaussian constants); see Table 4. The residual gap in the last column is the min–max slack of the $\sigma_{\min}$ sandwich (7), which the single-coordinate prediction does not resolve; it is bounded and of the observed size.

(3) Fitting a slope in $\log n$ over $n \le 14$ gives -0.58 rather than -1; this is a finite-size effect, not a discrepancy.

(4) The same deterministic prediction, evaluated for large n, converges to $-\log n + 1.12 + o(1)$: the constant inside the theory converges only logarithmically (its per-coordinate part is now closed-form, Proposition 8.4), and at $n = 14$ about 40% of it is still missing. The theorem's coefficient $-\frac{d(d-1)}{2} = -1$ is thereby confirmed by the $n \to \infty$ behaviour of the exact integrals, while the finite-n data sit on the exact finite-n curve.

	$\mathbb{E}_{\text{roots}} \log \lvert\det J\rvert$		$\mathbb{E}_{\text{roots}} \log \|v\|\|w\|$		$\mathbb{E}_{\text{roots}} \log(\kappa_{\text{Cay}}/\kappa_{\text{root}})$	
n	pred.	meas.	pred.	meas.	pred.	meas.
4	5.39	5.39	1.93	1.90	-1.18	-1.41
8	8.41	8.42	3.06	3.11	-1.55	-1.67
14	11.11	11.11	4.04	3.99	-1.93	-2.16

Table 4: Measured root averages at $d = 2$ for the Cayley method against the finite-n deterministic predictions of the theory.

Five further measurements were carried out with the same infrastructure.

Cayley at $d = 3$. The trivariate Cayley tensor was built by convolution and two Bézout-type divisions, and self-tested: the identity of Lemma 10.1 holds to 10^{-11} in the log across all roots, and all $6n^3$ roots are recovered at $n = 2, 3, 4$, validating the construction. The measured ratio $\mathbb{E}_{\text{roots}} \log(\kappa_{\text{Cay}}/\kappa_{\text{root}})$ is $-2.53, -3.00, -3.29$, negative and monotonically decreasing, consistent in sign and direction with the coefficient $-\frac{d(d-1)}{2} = -3$ of Theorem 10.1, with the strong finite-size attenuation expected at such small n.

Sylvester. The identity of Lemma 10.1 for the Sylvester matrix holds to relative error below $4 \cdot 10^{-3}$ in the log (degrading with the conditioning of $S(x_2^*)$ at outer roots, but negligible against the means), and the measured $\mathbb{E}_{\text{roots}} \log(\kappa_{\text{Sylv}}/\kappa_{\text{root}})$ is positive and increasing, $0.73, 0.98, 1.19, 1.38$ at $n = 4, 6, 8, 11$, and is reproduced within 0.06–0.12 by the finite-n

prediction built from the conditional fiber moment G_c of Section 10.2, whose asymptotic coefficient is $+\frac{1}{2}$.

Coordinate pair energy. The coordinate pair energy $\frac{1}{N}\sum_j \sum_{i\neq j} \log|x_{2,j} - x_{2,i}|$, the structural term of Gröbner elimination, measures $9.9, 17.4, 29.0, 46.2$ at the same n, against the independent-pair quadrature prediction $9.3, 17.4, 26.6, 42.2$: correlations contribute under ten percent, and the growth is $\Theta(n \log n)$, an outer-root effect again, since $\mathbb{E}\log|x - x'|$ for an independent pair equals $\mathbb{E}\log\max(|x|, |x'|) \approx (\log n)/n$ per pair, times $N - 1 = 2n^2 - 1$ pairs.

Projected pair energy. The projected pair energy of the rational univariate representation with balanced u: measured $11.1, 19.9, 31.4, 50.4$ at $n = 4, 6, 8, 11$, against the one-point Monte Carlo $(N-1)I_n = 10.5, 19.6, 29.1, 48.1$, with $I_n \to I(u) = 7\zeta(3)/(2\pi^2) - \frac{1}{2}\log 2$ (Proposition 10.2); correlations again contribute under five percent.

Overlap term. The overlap term Θ_n at $d = 1$ and $d = 2$; this is reported in Section 9.5, where it decides between the two candidate regimes. The scripts are short (a Bézout builder, a companion linearization, Newton refinement, and quadrature routines) and are available alongside this paper.

10.4 Relative conditioning: resolution of the apparent paradox

Theorem 10.1 says that in the absolute metric the Cayley eigenvalue is typically *better* conditioned than the root. This is not absurd, but it is a statement about units: an absolute perturbation of size ε means something different for the resultant matrix, whose entries are degree-d forms in the coefficients and are large, than for the polynomials themselves. Following [18, Sec. 5], the relative condition numbers are $\kappa_{\mathrm{rel}} = \kappa_{\mathrm{abs}} \cdot \|D\|/\|X\|$ with D the data and X the solution, so

$$\mathbb{E}_{\mathrm{roots}} \log \frac{\kappa_{\mathrm{rel}}(x_d, R_{\mathrm{Cay}})}{\kappa_{\mathrm{rel,root}}} = \mathbb{E}_{\mathrm{roots}} \log \frac{\kappa(x_d, R_{\mathrm{Cay}})}{\kappa_{\mathrm{root}}} + \mathbb{E}\log \sup_{|x_d|\leq 1} \|R_{\mathrm{Cay}}(x_d)\|_2 - \mathbb{E}\log \max_i \sup_\Omega |p_i| + O(1),$$

the $O(1)$ collecting $\mathbb{E}\log(\|x^*\|/|x_d^*|)$.

The data scales are Gaussian quantities. For the polynomials, $\mathbb{E}\log\sup_\Omega|p_i| = \frac{d}{2}\log n + O(\log(dn))$, since the field variance on the torus is $(n+1)^d$. For the resultant matrix, the Frobenius norm of the coefficient tensor A carries a near-diagonal enhancement that a naive Parseval count misses: at $d = 2$, Parseval in all three variables and the exact two-point formula

$$\mathbb{E}|f_{\mathrm{Cay}}(s,t,x)|^2 = \frac{2\,(n+1)^2\big[(n+1)^2 - |k_n(s\bar{t})|^2\big]}{|s-t|^2}$$

on the torus give $\mathbb{E}\|A\|_F^2 \asymp n^5$: the Dirichlet-kernel integral over $s\bar{t}$ contributes a factor n^3, not n^2. In general each eliminated pair contributes one extra factor of n, so $\mathbb{E}\|A\|_F^2 \asymp_d n^{d^2+d-1}$.

The operator norm behaves differently, and this is the key point, which can be closed for every d.

Proposition 10.1 *For every fixed d, $\mathbb{E}\log\sup_{|x|=1}\|R_{\mathrm{Cay}}(x)\|_2 = \frac{d^2}{2}\log n + O_d(\log\log n)$, and the coefficient of $\log n$ in the average relative Cayley ratio is 0.*

Proof sketch. Upper bound, by induction on d. Expanding the Cayley determinant along its first row gives $f_d = \sum_j \Delta_{s_1,t_1}(p_j)\cdot g_j$, where Δ_{s_1,t_1} is the divided difference in the first eliminated pair and g_j is the Cayley function of the j-th $(d-1)\times(d-1)$ minor, which is divisible by each remaining $(s_i - t_i)$ individually because its defining rows coincide there. Hence $R_d(x_d) = \sum_j H(\Delta p_j)\circ R^{(j)}_{d-1}$, a sum of compositions of triangular Hankel–Toeplitz sections whose symbols are the polynomials on subtori and lower-order Cayley matrices; at $d = 2$ this is the classical Gohberg–Semencul-type factorization of the Bézout matrix. Each finite triangular section of a bounded symbol costs a factor $C\log n$ (finite-section Riesz projection), and by Salem–Zygmund with Gaussian concentration $\mathbb{E}\log\sup_{\mathbb{T}^d}|p_j| = \frac{d}{2}\log n + O(\log\log n)$; unwinding the induction, $\log\sup_x\|R_d(x)\|_2 \le \sum_{j=1}^{d}\log\sup|p_j| + O_d(\log\log n) = \frac{d^2}{2}\log n + O_d(\log\log n)$ in expectation. Lower bound: $\|R(x)\|_2 \ge \|R(x)\|_F/\sqrt{(d-1)!\,n^{d-1}}$, with $\mathbb{E}\|R(x)\|_F^2 \asymp_d n^{d^2+d-1}$ (exact at $d = 2$ by the two-point formula above, one near-diagonal factor n per eliminated pair in general, at the sketch level already flagged), and the lower deviation of the degree-d Gaussian chaos $\|R(x)\|_F^2$ is controlled by Carbery–Wright, so $\mathbb{E}\log\|R(x)\|_2 \ge \frac{d^2}{2}\log n - O_d(1)$ at any fixed torus point. □

The spectrum of $R(x)$ on the torus is flat: with rank $(d-1)!\,n^{d-1}$, the flat-spectrum operator norm is

$$n^{\frac{d^2+d-1}{2}-\frac{d-1}{2}} = n^{\frac{d^2}{2}},$$

the near-diagonal enhancement of the Frobenius norm exactly compensating the rank normalization. The measured exponents at $d = 2$ confirm both statements: the fitted slope of $\log\sup_{|x|=1}\|R(x)\|_2$ over $n \le 24$ is 2.05 against $d^2/2 = 2$, while $\log\|R(x)\|_F$ and $\log\|A\|_F$ fit 2.4 and drift toward $(d^2+1)/2 = 2.5$. A rigorous lower bound consistent with this comes from testing against the monomial vectors v_s, v_t of the two groups of Cayley variables, for which $f_{\mathrm{Cay}}(s,t,x) = v_s^T R(x)\,v_t$; the matching upper bound is the content of Proposition 10.1.

Taking $\mathbb{E}\log\sup_x\|R(x)\|_2 = \frac{d^2}{2}\log n + O_d(1)$, the coefficient of $\log n$ in the average relative ratio is

$$-\frac{d(d-1)}{2} + \frac{d^2}{2} - \frac{d}{2} = 0:$$

relative neutrality. This is now supported on two independent fronts: the direct measurement of the relative ratio in Section 10.3, which increases toward and then settles near 0.6 in natural-log units over the accessible range, and the measured operator-norm exponent above. The picture is coherent: absolutely, the Cayley eigenvalue is typically better conditioned than the root by the polynomial factor of Theorem 10.1; relatively, the two are typically comparable; and the exponential worst cases of [18] are tail events in both metrics.

10.5 The general factorization, for other resultant matrices

Lemma 10.1 is specific to the Cayley and Sylvester matrices with their canonical eigenvector normalizations. For other resultant matrices, or other weightings, the following general

factorization applies; it is superseded by Lemma 10.1 in the two canonical cases, through the identity $\gamma_j = r'(x_j)/\det J(z_j)$ noted above. Let $S(x)$ be the $2n \times 2n$ Sylvester matrix polynomial in the y-coefficients of $p_1(x,\cdot), p_2(x,\cdot)$; $r(x) := \det S(x) = \mathrm{Res}_y(p_1, p_2)$ has degree $N = 2n^2$ and its roots are the x-coordinates of the roots $z_j = (x_j, y_j)$. The method computes the x_j as eigenvalues of the polynomial eigenvalue problem $S(x)v = 0$. With weights $\omega_0, \ldots, \omega_n$ on the matrix coefficients, the normwise unstructured condition number of the eigenvalue x_j is [21]

$$\kappa_{\mathrm{res}}(x_j) = \frac{\left(\sum_{k=0}^{n} \omega_k |x_j|^k\right) \|w_j\|_2 \, \|v_j\|_2}{|w_j^* \, S'(x_j) \, v_j|}, \tag{10}$$

with v_j, w_j right and left null vectors of $S(x_j)$. Unstructured is the honest choice: the QZ iteration perturbs the matrices, not the original coefficients, and the structured–unstructured gap is part of the instability mechanism of [18].

Lemma 10.2 *Let x_j be a simple root of $r = \det S$ and v_j, w_j null vectors of $S(x_j)$, in any normalization. Then* $\operatorname{adj} S(x_j) = \gamma_j v_j w_j^*$ *for a unique $\gamma_j \neq 0$, and:*

(a) $r'(x_j) = \gamma_j \, w_j^* S'(x_j) v_j$*;*

(b) $\gamma_j \, (w_j^* v_j) = \prod_{\mu \in \operatorname{spec} S(x_j) \setminus \{0\}} \mu$*, the deflated determinant of $S(x_j)$;*

(c) combining (a) and (10)*,*

$$\kappa_{\mathrm{res}}(x_j) = \Big(\sum_{k=0}^{n} \omega_k |x_j|^k\Big) \frac{\|v_j\| \, \|w_j\| \, |\gamma_j|}{|r'(x_j)|}, \qquad r'(x_j) = \mathrm{lc}(r) \prod_{i \neq j} (x_j - x_i),$$

and the right-hand side is invariant under separate rescaling of v_j and w_j.

Proof. Simplicity gives $\operatorname{rank} S(x_j) = 2n - 1$, so $\operatorname{adj} S(x_j)$ has rank one; from $S \cdot \operatorname{adj} S = \operatorname{adj} S \cdot S = 0$ at x_j, its column space is $\ker S(x_j)$ and its row space the left kernel, whence $\operatorname{adj} S(x_j) = \gamma_j v_j w_j^*$. (a) is Jacobi's formula $r'(x) = \operatorname{trace}(\operatorname{adj} S(x) \, S'(x))$ evaluated at x_j. (b): for any matrix A with a simple eigenvalue 0 and null vectors v, w, one checks on a Schur form that $\operatorname{adj} A = \big(\prod_{\mu \neq 0} \mu\big) \frac{v w^*}{w^* v}$; comparison gives (b). (c): substitute (a) into (10); rescaling $v_j \mapsto c v_j$ forces $\gamma_j \mapsto \gamma_j / c$, leaving $\|v_j\| \|w_j\| |\gamma_j|$ unchanged, and similarly for w_j. □

Each factor of $\kappa_{\mathrm{res}} / \kappa_{\mathrm{root}} = \sigma_{\min}(J) \, \kappa_{\mathrm{res}}$ sits at a known level:

(i) $\sigma_{\min}(J)$ and the weight $\sum \omega_k |x_j|^k$ are local, computed by Proposition 8.3 and the argument used for C_n.

(ii) $\|v_j\|$ is local: for Sylvester, $v_j = (y_j^{2n-1}, \ldots, y_j, 1)^T$, the monomial vector of the other coordinate; its log-average is the $d = 1$ functional of Section 8.2 with degree $2n - 1$, outer-root inflation included, hence $\Theta(\log n)$ with a constant delivered by the computation behind (8).

(iii) $\|w_j\|$ is local on the fiber: $w_j^* S(x_j) = 0$ encodes the cofactor identity $u(y)p_1(x_j, y) + t(y)p_2(x_j, y) \equiv 0$ with $u \propto p_2(x_j, y)/(y - y_j)$ and $t \propto -p_1(x_j, y)/(y - y_j)$. Thus $\|w_j\|$ depends only on the two univariate fiber polynomials at $x = x_j$, conditioned to share the root y_j; this is a Gaussian conditioning on the fiber, within the reach of the same pipeline.

(iv) $|r'(x_j)|$ is a pair statistic:

$$\frac{1}{N}\,\mathbb{E}\sum_j \log|r'(x_j)| = \mathbb{E}\log|\operatorname{lc}(r)| + \frac{1}{N}\,\mathbb{E}\sum_j\sum_{i\neq j}\log|x_j - x_i|,$$

the logarithmic pair energy of the x-marginal of the root set, computed at leading order in Section 10.3. For the canonical Sylvester and Cayley cases this term cancels against $|\gamma_j|$ by Lemma 10.1; it survives for the scalar-eliminant methods of Section 10.6.

(v) $|\gamma_j|$ is the deflated determinant of $S(x_j)$; in the canonical normalization it equals $|r'(x_j)|/|\det J(z_j)|$, so it needs no separate analysis there.

10.6 Gröbner elimination and the rational univariate representation

Both methods pass through an intermediate univariate polynomial, and [16] measures their instability by the conditioning $\kappa_{\text{uni}} = 1/|g'|$ of its roots.

For Gröbner elimination in the variable x_1, the eliminant g generates $I \cap \mathbb{C}[x_1]$; generically $\deg g = N$, its roots are the x_1-coordinates, and, for a generic system, g agrees with $\operatorname{Res}_{x_2,\dots}(p)$ up to a constant. With the monic normalization $g(x) = \prod_i (x - x_{1,i})$,

$$\kappa_{\text{uni}}(x_{1,j})^{-1} = |g'(x_{1,j})| = \prod_{i\neq j}|x_{1,j} - x_{1,i}|,$$

so

$$\frac{1}{N}\,\mathbb{E}\sum_j \log\frac{\kappa_{\text{uni}}(x_{1,j})}{\kappa_{\text{root}}(z_j)} = A_n - \frac{1}{N}\,\mathbb{E}\sum_j\sum_{i\neq j}\log|x_{1,j} - x_{1,i}|\,:$$

the structural term of Gröbner elimination is the coordinate pair energy. Note the contrast with Lemma 10.1: the matrix formulation of elimination is local, the scalar one is not. The normalization of g matters for κ_{uni}, and monic is the natural theoretical choice, matching the examples of [16]; other scalings contribute terms of the form $\mathbb{E}\log|\operatorname{lc}|$, controlled as above. For contrast, the worst case of this method in [16, Ex. 3.1] is σ^{-2^d+2}, while the average is a pair energy of N points concentrated near a circle, which Section 10.3 shows to be $2n(\log n + C)(1 + o(1))$ at $d = 2$: the monic eliminant has exponentially large derivatives at its roots, so in the absolute metric elimination too is typically *better* conditioned than the root, one more units effect; the relative version needs the coefficient scale of the eliminant, a Carbery–Wright quantity.

For the rational univariate representation, one projects by a linear form $t(x) = u^T x$ with $\|u\| \le 1$ and solves $f(x) = \prod_i (x - t(z_i))$, which is monic, so

$$\kappa_{\mathrm{uni}}(t(z_j))^{-1} = \prod_{i \neq j} |t(z_j) - t(z_i)| :$$

the pair energy of a one-dimensional linear projection of the whole root set. This energy can be computed at leading order, with a closed-form constant in the balanced case.

Proposition 10.2 *Let $d = 2$, $t(x) = u_1 x_1 + u_2 x_2$ with $u_1, u_2 > 0$, $u_1^2 + u_2^2 = 1$. Then*

$$\frac{1}{N}\,\mathbb{E} \sum_j \sum_{i \neq j} \log |t(z_j) - t(z_i)| = I(u)\, N\,(1 + o(1)), \qquad I(u) = \mathbb{E}\big[\log \max(u_1 D,\, u_2 D')\big],$$

where D, D' are independent copies of $2|\sin(\beta/2)|$ with β uniform on $(0, 2\pi)$, the correlation correction being measured below five percent at the accessible sizes, a constant-level bookkeeping step left at the flagged-sketch level. In the balanced case $u_1 = u_2 = 1/\sqrt{2}$,

$$I = \frac{7\,\zeta(3)}{2\pi^2} - \frac{\log 2}{2} = 0.0797\ldots > 0.$$

Proof. At leading order the pairs decouple and, by Proposition 8.1, the two coordinates of a single root become independent with uniform phases on the unit circle. Writing $e^{i\varphi} - e^{i\varphi'} = e^{i\alpha} D$ with α uniform and independent of $D = 2|\sin((\varphi - \varphi')/2)|$, the difference of two independent projected values is $u_1 D e^{i\alpha} + u_2 D' e^{i\alpha'}$. The mean-value property of the harmonic function $\log|\cdot|$ gives $\mathbb{E}_\alpha \log|m e^{i\alpha} + c| = \log \max(m, |c|)$ for any fixed c, and averaging over one phase yields $I(u)$. In the balanced case, with $F_D(2\sin y) = \frac{2}{\pi}\min(y, \pi - y)$,

$$\mathbb{E} \log \max(D, D') = \frac{8}{\pi^2} \int_0^{\pi/2} y\, \log(2 \sin y)\, dy = \frac{7\,\zeta(3)}{2\pi^2},$$

by the classical log-sine evaluation $\int_0^{\pi/2} y \log(2\sin y)\, dy = \frac{7}{16}\zeta(3)$, and $\mathbb{E}\log(D/\sqrt{2}) = -\frac{1}{2}\log 2$ since $\mathbb{E}\log D = 0$. □

The sign is noteworthy: $I > 0$ in the balanced case, so the monic eliminant of the rational univariate representation has *exponentially large* derivatives at its roots, and in the absolute metric κ_{uni} is exponentially small on average, the most extreme units effect in the family. Near-coincidences of projected roots, the mechanism of the devastating example of [16, Sec. 4], are integrated inside $I(u)$ and only shape the constant; the example itself is a tail event. The finite-n version of the statement, a one-point Monte Carlo of $(N-1)I_n$, matches the measured energies within five percent (Section 10.3). For general u the one-dimensional integral defining $I(u)$ is immediate, and for general d the same harmonic reduction applies to the outermost phase, leaving a $(d-1)$-fold integral.

10.7 Multiparameter eigenvalue problems: local, computed for an explicit representation

The multiparameter approach represents each p_k as the determinant of a matrix of affine-linear forms, $W_k(x) = A_{k,0} + \sum_l x_l A_{k,l}$ with $\det W_k = p_k$, and solves the resulting multiparameter eigenvalue problem through its operator determinants [16]. For the representation to be computable directly from the coefficients one needs a *uniform* determinantal representation in the sense of [7], with entries affine-linear in the c_α as well; the minimal size of such a representation is $\Theta(n^{d/2})$ for fixed d, and for $d = 2$ the explicit MinUnif construction of [7, Ex. 5.1] has size $2n+1$: $W = \left[\begin{smallmatrix} M_x & 0 \\ L & M_y^T \end{smallmatrix}\right]$, with M_x, M_y bidiagonal in $(-x, 1)$ and $(-y, 1)$, and the coefficients sitting, each exactly once, in the antitriangular block L. The first observation is that the method is local, for *every* uniform representation, by the adjugate identity.

Proposition 10.3 *Let z^* be a simple root and, for each k, let $W_k(z^*)$ have a one-dimensional kernel; normalize the null vectors by* $\operatorname{adj}(W_k(z^*)) = v_k w_k^T$. *Then:*

(a) $\prod_j' \sigma_j(W_k(z^*)) = \|v_k\|\,\|w_k\|$ *and* $J_{kl} = \partial_l p_k(z^*) = w_k^T A_{k,l} v_k$*: the structural content of the multiparameter method is* $\sum_k \log(\|v_k\|\|w_k\|) - \log|\det J|$*, of the same form as for the resultant methods (Lemma 10.1), and in particular local;*

(b) for MinUnif *in two variables the null vectors are explicit:* $v_k = (1, x, \dots, x^n;\, b^{(k,y)})$ *and* $w_k = (b^{(k,x)};\, 1, y, \dots, y^n)$*, where $b^{(k,y)}$ collects the Horner shifts of the fiber $p_k(x, \cdot)$ deflated at the root, and symmetrically for $b^{(k,x)}$: the conditioned deflated-fiber objects of the Sylvester analysis, now with the triangular variance profile $V_j = k_{n-j}(r)$, for which the rank-one Schur complement generalizes verbatim:*

$$\mathbb{E}_{\mathrm{cond}}\|b\|^2 = \sum_k \frac{\left[\sum_{j>k} V_j r^{j-k-1}\right]\left[\sum_{j\le k} V_j r^j\right]}{\sum_j V_j r^j}\,;$$

(c) in the total-degree model at $d = 2$, $\mathbb{E}_{\mathrm{roots}} \log\|v_k\| = \frac{5}{2}\log n + O(1)$, the window contributing $\frac{3}{2}\log n$ and the collective radial excursions of Proposition 11.2(c) contributing $\log n$; hence

$$\frac{1}{N}\,\mathbb{E}\sum_j \log\frac{\kappa_{\mathrm{MEP}}(z_j)}{\kappa_{\mathrm{root}}(z_j)} = (d+1)\log n + 4\log n + O(1) = 7\log n + O(1),$$

the collective-tail step being at the same sketch level as in Proposition 11.2.

Proof sketch. (a) At a simple singular point $\operatorname{adj}(W)$ has rank one with $\|\operatorname{adj}(W)\|_2 = \prod_j' \sigma_j$, and Jacobi's formula $\partial_l \det W = \operatorname{trace}(\operatorname{adj} W\, \partial_l W)$ gives the gradient identity, $\partial_l W_k = A_{k,l}$ being constant. (b) is back-substitution in the two bidiagonal blocks, the right-hand sides being the fiber sections $(Lv^{(1)})_i = a_{i-1}(x)$; the Schur computation is the Sylvester one with V_j in place of a constant variance. (c) On the window the profile sums give $\mathbb{E}_{\mathrm{cond}}\|b\|^2 \asymp n^3$; on a collective excursion of height t the products $V_j r^j$ are flat in j and the conditioned sum

is $\asymp e^t(n/t)^2$, so $\log\|v\|^2 \approx t + 2\log n - 2\log t$ there, and integrating the excess against the tail d/t of Proposition 11.2(c) adds $2\log n$ to the window value $3\log n$; the determinant and $\sigma_{\min}$ averages are those of Proposition 11.2(d). □

Numerically, at total-degree roots with the pipeline of Section 10.3: the MinUnif determinant identity holds at machine precision; the adjugate identity of (a) holds to 10^{-3} in the log at $n = 11$, degrading with the conditioning of W_k at far roots but negligible against the means; and the measured structural term and full ratio are $3.2, 3.5, 4.2, 4.7$ and $4.7, 5.7, 7.1, 8.1$ at $n = 4, 6, 8, 11$, with fitted slopes 1.5 and 3.5 and rising increments, the finite-size attenuation matching that of the certified local averages of the model at equal sizes. Two remarks are in order. First, the structural term depends on the chosen representation: smaller or better-scaled uniform representations change the constant, and possibly the coefficient, so the representation is a genuine design lever for the average-case conditioning of the method. Second, the qualitative conclusion stands regardless: in the absolute metric the multiparameter subproblem is, on average, polynomially worse conditioned than the root, and the exponential worst cases are tail events here as everywhere else in the family.

10.8 Macaulay resultant: back to the overlap class

By [16, Thm. 7.1], the Macaulay eigenvalue problem built with a random linear h satisfies

$$\kappa_{\mathrm{eig}}(\lambda^*) \;\geq\; \frac{\|[\det Q]_{\mathcal{B}}\|_2\, \|V_{\mathrm{col}}(x^*)\|_2}{|\det J(x^*)\, h(x^*)|},$$

where $\mathcal{B}$ is the basis given by the retained h-rows and $V_{\mathrm{col}}(x^*)$ is the vector of column labels (monomials of degree at most $\rho := \sum_i \deg p_i - d + 1$) evaluated at the root. Theorem 3.1 applies verbatim: $\|[\det Q]_{\mathcal{B}}\| = |\det J|\, \|[\ell_j]_{\mathcal{B}}\|$, so

$$\kappa_{\mathrm{eig}}(\lambda^*) \;\geq\; \frac{\|[\ell_j]_{\mathcal{B}}\|\, \|V_{\mathrm{col}}(x^*)\|}{|h(x^*)|}.$$

The Macaulay method therefore sits in the overlap class together with the normal form: the same dual norm $\|[\ell_j]_{\mathcal{B}}\| = \|V^{-T}e_j\|$, with the identities of Lemma 9.1, paired with a longer but local monomial vector (Section 8.2 machinery with staircase degrees up to ρ) and the harmless local factor $|h(x^*)|^{-1}$. The two-dimensional refinement of [16, Sec. 8], which bounds the numerator below by $\sigma_{\min}$ of the Macaulay matrix, offers a complementary handle: that smallest singular value is again controlled in expectation by Carbery–Wright, and is a natural companion statistic for the numerical protocol.

10.9 The hierarchy

Table 5 collects the classification established above, together with the average-case result now available at each level; it doubles as the scoreboard of this paper.

Method	structural object	class	absolute	relative
Cayley resultant	null-vector norms, $\det J$	local	$-\frac{d(d-1)}{2}\log n$ (Thm. 10.1)	0
Sylvester ($d=2$)	conditional fiber moment	local	$+\frac{1}{2}\log n$ (§10.2)	$+\frac{1}{2}\log n$
Multiparameter EP	deflated dets of $W_k(x^*)$	local	$+7\log n$ at $d=2$ (§10.7)	$+\frac{13}{2}\log n$
Gröbner elimination	coordinate pair energy	pairwise	$\approx -2n\log n$ at $d=2$	$\approx -n(\log n + C)$
Rational univariate	projected pair energy	pairwise	$-I(u)\,N$ (Prop. 10.2)	$+\delta N$
Normal form	overlap $\Theta_n(\mathcal{B})$	full config.	$(d{+}1)\log n{+}O(1)$, $d\le 2$ (Thm. 9.1)	$(\frac{d}{2}{+}1)\log n{+}\mathbb{E}\log\|M_g\|$
Macaulay resultant	overlap $\|[\ell_j]_{\mathcal{B}}\|\,\|V_{\mathrm{col}}\|$	full config.	lower bounds (§10.8)	lower bounds

Table 5: Summary of the average-case results, per method: the structural object, its class, and the leading behaviour of the average logarithmic conditioning ratio in the absolute metric. Pairwise entries are stated at $d = 2$; the relative column follows the conversion of Section 12.

The local terms are common to all methods, so any average-case comparison between methods reduces to comparing structural objects. And the ordering by average-case tractability does not follow the worst-case results of [16], where all methods are exponentially unstable; the local class already contains a complete theorem (Theorem 10.1), and the pairwise class requires only two-point information.

10.10 Basis choices within the normal-form family

The normal-form variants of [16, 20] correspond to different quotient bases $\mathcal{B}$ and different g. Theorem 3.1 and Corollary 3.1 hold for any basis, with $\Lambda_j = \Lambda_j(\mathcal{B})$, and the local terms are basis-independent. Hence the average-case comparison between variants is the comparison of the overlap averages $\Theta_n(\mathcal{B})$, and the protocol of Section 9.5 should be run with one curve per basis, on the same sampled systems; the vertical offsets between the curves are the average-case ranking.

11 The normal-form theorem and interpretation

Theorem 11.1 (normal-form method; conditional in its last claim) *In the model of Definition 2.1, for fixed d and $n \to \infty$,*

$$\mathcal{R}_n = (d+1)\log n + \Theta_n + O(d), \qquad 0 \le \Theta_n < \infty,$$

with $\Theta_n < \infty$ by Proposition 9.4. In particular $\mathcal{R}_n \ge (d+1)\log n - O(d)$. By Theorem 9.1, unconditionally for $d = 1, 2$ (and, at $d \ge 3$, whenever $\Theta_n = O(d)$),

$$\mathcal{R}_n = (d+1)\log n + O(d) :$$

the local terms are the whole story.

Proof. Combine Corollary 3.1, so that $\mathcal{R}_n = A_n + \Theta_n + C_n$ as an identity of finite quantities by Lemma 6.1 and Proposition 9.4, with Proposition 8.3, under the conjecture insert $\Theta_n = O_d(1)$. □

REMARK **11.1** *Two readings. First, average-case tameness: the average logarithmic loss grows logarithmically in the Bézout count, so there is no exponential average-case penalty in this model. The worst-case instability of [18, 16] lives in the upper tail of Λ, produced by nearly colliding roots, whose contribution Section 9.4 shows to be integrable, and in adversarial geometries. This agrees with the histograms in our earlier notes. Second, a caveat on the yardstick: κ_{eig} is invariant under $P \mapsto cP$ while κ_{root} is not, and $\sigma_{\min}(J)$ is exponentially large at outer roots, where the absolute root conditioning is spuriously excellent. A substantial part of the deterministic term $(d+1)\log n$ is this mismatch of units. A scale-free comparison removes it:*

$$\mathbb{E}_{\mathrm{roots}} \log\Big[\kappa_{\mathrm{eig}} \cdot \frac{\sigma_{\min}(J)}{\sqrt{S\, h_\bullet}}\Big] = \Theta_n + C_n + \mathbb{E}_W[\log \sigma_{\min}(G)] + O(\log d),$$

whose deterministic part is $O(\log d)$. In this normalization the entire n-dependence of the average ratio is Θ_n, which is bounded for $d \le 2$ by Theorem 9.1.

11.1 The Kostlan model: an extreme units effect

In the Kostlan model ($p_i = \sum_{|\alpha|\le n} c_{i,\alpha}\sqrt{\binom{n}{\alpha}}\, x^\alpha$ with i.i.d. $c_{i,\alpha} \sim \mathbb{CN}(0,1)$, kernel $K(x,y) = (1+\langle x, y\rangle)^n$, Bézout count $N = n^d$), Proposition 4.2 shows that the root-average measure does not depend on n. This changes the geometry of the problem completely, and the local terms can be computed in closed form.

Proposition 11.1 *Under the root average of the Kostlan model, with $s := \|X\|^2$ and $H_d := \sum_{k=1}^d \frac{1}{k}$:*

(a) $u := s/(1+s)$ has density $d\, u^{d-1}$ on $(0,1)$, and $\mathbb{E}_{\mathrm{roots}}[\log(1+s)] = H_d$, exactly;

(b) $\mathbb{E}_{\mathrm{roots}}[\log \sigma_{\min}(J)] = \frac{H_d}{2}\, n + \frac{1}{2}\log n + O_d(1)$ and $\mathbb{E}_{\mathrm{roots}}[\log|\det J|] = \frac{dH_d}{2}\, n + \frac{d}{2}\log n + O_d(1)$;

(c) the normal-form conditioning ratio satisfies

$$\mathcal{R}_n^{\mathrm{Kos}} = \frac{H_d}{2}\, n + O(\log n) + \Theta_n^{\mathrm{Kos}}.$$

Proof. (a) The normalized root density is $\frac{d!}{\pi^d}(1+s)^{-(d+1)}$, so the radial variable has density $d\, s^{d-1}(1+s)^{-(d+1)}$; the substitution $u = s/(1+s)$ gives $d\, u^{d-1}$, and $\mathbb{E}\log(1+s) = -d\int_0^1 u^{d-1}\log(1-u)\, du = H_d$, by the classical integral $\int_0^1 u^{k-1}\log(1-u)\, du = -H_k/k$. (b) Here $\Sigma(X) = SH$ has eigenvalues $n(1+s)^{n-1}$, with multiplicity $d-1$, and $n(1+s)^{n-2}$ (proof of Proposition 4.2), so the sandwich (7) gives $\log\sigma_{\min}(J) = \log\sigma_{\min}(G) + \frac{1}{2}\log n + \frac{n-1}{2}\log(1+s) + O(\log(1+s))$, and by (a) the slack has expectation $O(H_d)$; the tilt is handled by Corollary 7.1 exactly as in Proposition 8.3, and the integrability check of Lemma 6.2 holds with the n-free spatial density. For the determinant, $\det\Sigma = n^d(1+s)^{(n-1)d-1}$. (c)

Reduction I is model-independent, and $\mathbb{E}_{\text{roots}} \log(1 + |\lambda|) = O(1)$ since the one-coordinate marginal has an n-free density with $\mathbb{P}(|x_1| > t) = O(t^{-2})$. Numerical check at $d = 1$: the measured $\mathbb{E}_{\text{roots}} \log |p'|$ is 10.707, 21.008, 41.420 at $n = 20, 40, 80$, against the prediction $\frac{n}{2} + \frac{1}{2} \log n - 1 + \frac{\psi(2)}{2} = 10.709,\ 21.056,\ 41.402$, and the measured $\mathbb{E} \log(1 + |z|^2)$ is 1.001 against $H_1 = 1$. □

This computation exposes two structural facts. First, there is no outer-root phenomenon here in the sense of Section 8.2: the root measure does not concentrate on a torus, so a fixed fraction of the roots carries affine local scales of size $(1 + s)^{\Theta(n)}$, and the units mismatch, logarithmic in the separate-degree model, becomes *linear* in n. The affine yardstick and the Kostlan ensemble do not belong together: the ensemble is unitarily invariant on projective space, and its natural condition numbers are the projective ones of Shub and Smale [19].

Second, in the scale-free normalization of Remark 11.1 the spatial term disappears identically:

$$\mathbb{E}_{\text{roots}} \log\Big[\frac{\sigma_{\min}(J)}{\sqrt{\lambda_\bullet(\Sigma)}}\Big] = \mathbb{E}_W[\log \sigma_{\min}(G)] + O_d(1),$$

with no n-dependence at all, so the entire n-dependence of the scale-free Kostlan ratio is the overlap term Θ_n^{Kos}. The Kostlan model is therefore the clean laboratory for the overlap question, and running the protocol of Section 9.5 on Kostlan roots is a natural complement to the separate-degree runs.

11.2 The total-degree model: same coefficients, different mechanism

In the total-degree i.i.d. model, $p_i = \sum_{|\alpha| \le n} c_{i,\alpha} x^\alpha$ with i.i.d. $\mathbb{CN}(0, 1)$ coefficients, the kernel is $K(x, y) = F(t)$ with $t_l = x_l \bar{y}_l$ and $F(t) = \sum_{|\alpha| \le n} t^\alpha$, and $N = n^d$. The simplex Newton polytope changes the geometry of the root measure, but not the local coefficients.

Proposition 11.2 *In the total-degree model, with $s_l := n \log r_l$ and $s_0 := 0$:*

(a) uniformly on compacts, $\log F(e^{s/n}) = d \log n + \log \Phi(s) + o(1)$, where

$$\Phi(s) = \int_{\Delta_d} e^{\langle \theta, s \rangle}\, d\theta = \sum_{l=0}^{d} \frac{e^{s_l}}{\prod_{m \ne l}(s_l - s_m)}$$

is the exponential divided difference on the nodes $\{0, s_1, \dots, s_d\}$, with $\Phi(0) = 1/d!$;

(b) the rescaled log-radii of a root converge in law to the Monge–Ampère density

$$m_{\text{td}}(s) = d!\, \det D_s^2 \log \Phi(s),$$

a probability measure: its mass is $d! \operatorname{Vol}\big(\nabla \log \Phi(\mathbb{R}^d)\big) = d! \operatorname{Vol}(\operatorname{int} \Delta_d) = 1$, since $\nabla \log \Phi$ is the moment map of the exponential family with uniform base measure on Δ_d; at $d = 1$ it reduces to the density m of Proposition 8.1;

(c) the heavy tail is collective: $\mathbb{P}(\min_l s_l > t) = \frac{d}{t}\,(1 + o(1))$*, while excursions of a single coordinate with the others bounded have probability* $O(t^{1-2d})$*; the far roots recede radially in all coordinates at once, as the simplex polytope dictates;*

(d) as a result,

$$\mathbb{E}_{\text{roots}} \log S = 2d \log n + O_d(1), \qquad \mathbb{E}_{\text{roots}} \log \sigma_{\min}(J) = (d+1)\log n + O_d(1),$$
$$\mathbb{E}_{\text{roots}} \log |\det J| = d(d+1)\log n + O_d(1),$$

and $\mathcal{R}_n^{\text{td}} = (d+1)\log n + \Theta_n^{\text{td}} + O_d(1)$*: the local coefficients coincide with those of the separate-degree model.*

Proof sketch. (a) is a Riemann sum: $\sum_{\alpha \in n\Delta_d} e^{\langle \alpha, s\rangle/n} = n^d \int_{\Delta_d} e^{\langle \theta, s\rangle} d\theta\,(1 + o(1))$, and the divided-difference evaluation of the simplex integral is classical. For (b), with $t_l = e^{s_l/n}$ the diagonal part of $H_{jk} = \delta_{jk}\partial_{t_k} \log F + \bar{x}_j x_k\, \partial_j \partial_k \log F$ cancels at leading order against the chain-rule correction, leaving $H_{jk} = \bar{x}_j x_k\, n^2 (t_j t_k)^{-1} (D_s^2 \log \Phi)_{jk}(1 + o(1))$, so $\det H = n^{2d} \prod_l r_l^{-1} \det D_s^2 \log \Phi$, and the intensity of Lemma 4.2 transported to the s-variables is $d! \det D_s^2 \log \Phi$; the mass statement is the moment-map property. For (c), write $s = (t + c)\mathbf{1} + w$ with $\sum w_l = 0$: since the node $s_0 = 0$ does not shift, $\Phi = \frac{e^{t+c}}{t+c}\,\psi(w)\,(1 + O(1/t))$ with ψ the divided difference on the nodes w alone, so $\log \Phi$ is affine in the radial direction up to curvature $O(t^{-2})$, and the gradient image of $\{\min_l s_l > t\}$ is the slab $\{\sum \theta_l > 1 - 1/t\}$ of Δ_d, of volume $\approx \frac{d}{t\,d!}$; multiplying by $d!$ gives the tail d/t. The single-coordinate slab carries $\det D^2 \log \Phi = O(t^{-2d})$ over an $O(1)$ region. For (d), $\log S = d \log n + \log \Phi$ and $\log \Phi \approx t$ on the deep tail, so $\mathbb{E}[\log \Phi]_+ = \int \mathbb{P}(\log \Phi > t)\, dt = d \log n + O_d(1)$; the $\sigma_{\min}$ and determinant averages follow by the sandwich of Proposition 8.3, the deep-tail eigenvalue deficit ($\lambda_{\text{rad}} \sim t^{-2}$) integrating to $O_d(1)$ against the d/t tail. On the torus the eigenvalues of H are $\mu_{\tan} = \frac{n(n+d+1)}{(d+1)(d+2)}$, with multiplicity $d - 1$, and $\mu_{\text{rad}} = \mu_{\tan}/(d+1)$, so the eccentricity is exactly $d + 1$; at $d = 1$, $\mu = n(n+2)/12$, the Kac value, as it must be. Two steps are at sketch level: the uniformity in (a) and the tail-slab computation in (c). □

Numerical check at $d = 2$, with the usual pipeline on triangular coefficient supports (all n^2 roots recovered): the fitted slopes of $\mathbb{E} \log S$, $\mathbb{E} \log \sigma_{\min}(J)$, $\mathbb{E} \log |\det J|$ over $n \le 11$ are $2.6, 2.0, 4.2$ against the asymptotic $4, 3, 6$, with successive increments rising ($2.3 \to 2.9$, $1.8 \to 2.2$, $3.8 \to 4.7$), the same finite-size attenuation pattern shown by the separate-degree data at equal sizes, where the exact finite-n integrals certified the coefficients. The collective-tail diagnostic of (c) is clearly visible: the joint exceedance $\mathbb{P}(s_1 > 1 \text{ and } s_2 > 1)$ exceeds the product of the marginals by a stable factor ≈ 1.5, whereas in the separate-degree model the one-point intensity factorizes exactly across coordinates.

The three models now span the full picture. Separate degree: torus concentration, per-coordinate outer roots, local coefficients $(d+1)\log n$. Total degree: torus concentration, collective radial excursions, the *same* coefficients. Kostlan: no concentration, linear growth in n, projective normalization mandatory. The agreement of the two i.i.d. models strengthens the reading of Theorem 11.1: the coefficient $(d+1)\log n$ is a robust property of torus-concentrated Gaussian ensembles, not an artifact of the box supports.

12 The relative picture, and why it is the right one

Everything computed so far lives in the absolute metric of [16, 18], and deliberately so: it is the metric in which the Gaussian calculus is exact, every average is an identity or a quadrature, and the machinery of Sections 4–9 applies without normalizing anything. But it is not the metric in which conclusions should be drawn. Floating-point arithmetic commits *relative* errors; backward-stable eigensolvers return backward errors relative to the norm of the data; and the relative condition number is invariant under the rescalings of data and unknowns that every implementation performs, while the absolute one is not. In Gaussian ensembles this is not a pedantic distinction: the hidden scales are enormous and root-dependent, as the outer-root and units effects documented throughout this paper show, so absolute comparisons systematically conflate genuine sensitivity with the bookkeeping of scales. The absolute metric is the right vehicle for the computation; the relative metric is the right lens for the verdict. This section converts the entire scoreboard.

Following [18, Sec. 5], $\kappa_{\mathrm{rel}} = \kappa_{\mathrm{abs}} \cdot \|D\|/\|X\|$ with D the data of the subproblem and X its solution, so for every method

$$\widehat{\mathcal{R}}_M := \frac{1}{N}\,\mathbb{E}\sum_j \log\frac{\kappa_{\mathrm{rel}}(M)}{\kappa_{\mathrm{rel,root}}} = \mathcal{R}_M + \mathbb{E}\log\|D_M\| - \mathbb{E}\log\max_i \sup_\Omega |p_i| + O(1),$$

the output scales contributing $O(1)$ in these models, and the baseline being $\mathbb{E}\log\max_i \sup_\Omega |p_i| = \frac{d}{2}\log n + O(\log dn)$.

Proposition 12.1 (relative scoreboard) *In the separate-degree model, with the pairwise and local entries at $d = 2$:*

(a) Cayley: $\widehat{\mathcal{R}} = O_d(\log\log n)$ for every d, by Proposition 10.1: relative neutrality, the clean case.

(b) Sylvester: $\widehat{\mathcal{R}} = \frac{1}{2}\log n + O(\log\log n)$: the data scale $\sup_x \|S(x)\|_2$ coincides with $\sup|p|$ up to the Toeplitz-symbol log factors, so the relative and absolute coefficients agree.

(c) Multiparameter (MinUnif): $\widehat{\mathcal{R}} = \frac{13}{2}\log n + O(1)$, the data norm contributing $\frac{1}{2}\log n$ through the coefficient block.

(d) Normal form: $\widehat{\mathcal{R}} = (\frac{d}{2}+1)\log n + \mathbb{E}\log\|M_g(\mathcal{B})\| + O(1)$, basis-dependent through the norm of the multiplication matrix; measured, $\mathbb{E}\log\|M_g\| \approx \frac{1}{2}\log N$ for the companion matrix at $d = 1$ and $\approx 2\log n$ for the Bézout pencil at $d = 2$, giving $\approx 2\log n$ and $\approx 4\log n$ respectively.

(e) Gröbner elimination: $\widehat{\mathcal{R}} = -n(\log n + C)(1 + o(1))$: only half *of the pair energy cancels against $\mathbb{E}\log\|f\|$, because the coefficient norm of the eliminant sees one radial tail, $\sum_i \mathbb{E}\log\max(1, |x_i|)$, where the root-averaged potential sees two; measured, $\log\|f\|/\mathrm{PE} = 0.65, 0.64, 0.62$ at $n = 6, 8, 11$, trending to $\frac{1}{2}$.*

(f) Rational univariate representation: $\widehat{\mathcal{R}} = \delta\, N\,(1 + o(1))$ *with, in the balanced case,*

$$\delta = \frac{G}{\pi} + \frac{\log 2}{4} - \frac{7\,\zeta(3)}{2\pi^2} = 0.03857\ldots > 0,$$

G Catalan's constant: the unique exponential penalty of the family.

Proof sketch. (a) is Proposition 10.1 with the conversion formula. (b), (c), (d) insert the data scales computed or measured above; for (d) note that $\|M_g(\mathcal{B})\|$ is a property of the chosen basis, not of the quotient. For (e), $\mathbb{E}\log\|f\|$ per root is the circle average of the potential of the coordinate marginal, $\mathbb{E}\log\max(1,|x|) = \frac{1}{2n}(\log n + C')(1+o(1))$, a single-tail quantity, while the pair energy is the root average of the same potential, a two-tail quantity twice as large; the difference is $-n(\log n + C)$ at leading order. For (f), the same comparison for the projected marginal is a contest of constants: the circle potential of the limit law W is $c' = \mathbb{E}[\log_+|W|] = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2}\log(1+\cos\theta)\,d\theta = \frac{G}{\pi} - \frac{\log 2}{4}$, by the classical evaluation $\int_0^{\pi/2}\log(1+\cos\theta)\,d\theta = 2G - \frac{\pi}{2}\log 2$, while the pair energy constant is $I(u) = \frac{7\zeta(3)}{2\pi^2} - \frac{\log 2}{2}$ (Proposition 10.2); $\delta = c' - I(u)$. Measured, $(\log\|f\| - \mathrm{PE})/N = -0.073, -0.052, -0.029$ at $n = 6, 8, 11$, rising toward δ with the explicit $(\log n)/n$ outer transient common to both terms. □

The relative verdict is sharper than the absolute one, and closer to practical folklore. The resultant methods are benign: Cayley exactly neutral, Sylvester polynomially penalized, and the exponential worst cases of [18] are tail events in both metrics. The normal form pays a polynomial factor that depends on the basis through $\|M_g\|$, one more reason the basis is not a detail. Elimination remains distorted by units even relatively, now through the mismatch between the unit circle, where the coefficient norm of the eliminant is measured, and the region where its roots actually live. And the rational univariate representation is the one method the relative metric convicts: its projected root measure fails to be an equilibrium measure of the circle, its potential is not constant, and the resulting $e^{\delta N}$ average penalty is exactly the clustering mechanism of the devastating example of [16, Sec. 4] surviving into the average case, in the metric where it belongs. The practical reading: project only if you must, and if you must, the constant $\delta(u)$ is a design objective, minimized by making the projected measure as close to equilibrium as the geometry allows.

13 Conclusion

The results of this paper assemble into one picture. Every algebraic rootfinding method in the family of [16, 18] obeys the same average-case skeleton over Gaussian random systems: a universal local exchange rate between root and subproblem conditioning, computed here exactly and driven in part by the roots away from the unit torus, plus a structural term that identifies the method. The structural terms are now understood at every level except the sharpest: the Cayley method is solved unconditionally, with the conclusion that its exponential worst case is a tail event and that the comparison is neutral in the relative metric; the Sylvester, Gröbner and rational univariate terms are computed at leading order,

the last with a closed-form constant; and the full-configuration term of the normal form is proved bounded at $d = 1$ and 2, against the random-matrix calibration, so the normal-form theorem is unconditional there. The choice of coefficient model and of quotient basis is not a detail: each ensemble selects its own natural basis, and mismatching them converts logarithmic effects into linear ones.

For the record, the following are proved here: Theorem 3.1, Corollary 3.1, Lemmas 4.1, 4.2, 5.1, 6.1, 6.2, 7.1, 9.1, 9.2, 9.3 with Propositions 9.1 and 9.2 and Theorem 9.1, Lemma 10.1 (from [18]), Lemma 10.2, Propositions 4.1, 4.2, 8.1, 8.2 with Corollary 8.1 and the closed-form constant of Proposition 8.4, 8.3, 9.3 (fixed d), 9.4, 11.1 (with numerical check), 10.2 (with closed-form constant), 11.2 (sketch level in two steps, with numerical check), 10.3 (with numerical check); Theorem 10.1, verified numerically in Section 10.3 together with the relative analysis of Sections 10.4 and 12 (Proposition 12.1); the overlap term is resolved numerically at $d = 1$ and $d = 2$ in Section 9.5.

14 The open problem

Problem 14.1 *What remains of the overlap question. (i) Settle the regime at $d = 3$, where the measured overlaps run parallel to the Ginibre control, a stable* 0.2 *below it, up to $N =$* 1296*: extend the stratified second-moment route of Theorem 9.1, whose identity chain carries over to the trivariate Cayley pencil, or decide numerically at larger sizes. (ii) Identify the invariant basis for the Kostlan model at $d \geq 2$, where both the weighted box and the degrevlex Fubini–Study staircase fail.*

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